Incentivizing Work to Be Reviewed

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Abstract

How should an organization reward work that a manager subsequently uses and revises? An analyst's hidden effort determines the accuracy of an initial assessment. A manager combines a noisy observation of this assessment with independent state information to make the final decision. The organization chooses wages and purchases noisy records of the assessment, state, and decision. For any positive probability of high effort, a minimum-cost contract that sustains it in equilibrium rewards agreement among the three records. Holding this probability fixed, record prices determine how records adjust as the manager relies more on independent information. When the state record is sufficiently inexpensive to acquire exactly, optimal initial-record precision weakly rises. When the initial assessment is sufficiently inexpensive to record exactly, optimal state-record precision weakly falls. After wages, records, and effort are optimized using each contract's most profitable equilibrium, better managerial information weakly raises profit when initial recording is sufficiently inexpensive. Under other record prices, profit can fall locally even as effort stays high and decisions improve. Better information can also make the organization switch to low effort, reducing equilibrium accuracy.

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1 Introduction

Organizations often make decisions in stages. An analyst investigates a problem and produces an initial assessment. A manager combines that assessment with independent information to make the final decision. How much she relies on the assessment depends on its expected accuracy and on the strength of her independent evidence. Her use of the assessment affects both the quality of the final decision and the evidence available for evaluating the analyst's work.

Consider a resident who makes a preliminary diagnosis that an attending physician reviews before reaching a final diagnosis. When the attending adopts the diagnosis she receives, the final diagnosis preserves information about the resident's original judgment. When she draws on independent evidence, agreement between the two diagnoses can help check whether that judgment was correct. To evaluate the resident's work, the hospital can retain records of both diagnoses and obtain evidence about the patient's condition. How should an organization reward initial work that another decision maker subsequently uses and revises, and which records should it acquire for this purpose?

In the model, a principal hires an analyst to assess an unknown binary state. The analyst chooses unobserved effort. Low effort is costless; high effort is costly and produces a more accurate assessment. A manager receives a noisy observation of the assessment and an independent signal about the state. She chooses the final decision to maximize the probability of matching the state. Before effort, the principal commits to a wage schedule and to the precision of three records: the initial assessment, the state, and the final decision. These records become available after the decision. Each precision has a quadratic acquisition cost with its own price. The analyst is risk neutral and protected by limited liability. The principal values correct decisions and bears the expected wage and record acquisition costs.

The contract affects the accuracy the manager expects from the assessment; her response determines what the final decision reveals about the analyst's effort. A given contract can sustain several effort choices, each consistent with the manager's expectation. Under favorable evaluation, the principal values each contract at its most profitable equilibrium. For given record precisions, a minimum-cost wage supporting any positive probability of high effort pays a bonus when all three records agree. This event provides the greatest effort incentive per unit of expected payment. The principal then chooses record precisions to minimize expected wages plus acquisition cost and compares profits across effort probabilities. More precise evidence can reduce the rent required to sustain the same effort. Its acquisition is worthwhile when the wage saving covers the additional cost.

A more informative independent signal changes both the value of initial effort and the cost of purchasing it. Consider first contracts that sustain the same probability of high effort as the signal improves. When the state is sufficiently inexpensive to record exactly, the final decision helps reveal what the analyst originally concluded. As the manager relies more on independent evidence, the principal weakly increases initial-record precision to replace information previously supplied by the decision. When the initial assessment is sufficiently inexpensive to record exactly, the final decision helps check whether it was correct. Better independent evidence then allows the principal to weakly reduce state-record precision. In this case, the minimum cost of sustaining each effort probability weakly falls, and optimized profit under favorable evaluation weakly rises.

With a low state-record price, better managerial information can reduce profit even when high effort remains optimal. Cheap decision records can make it profitable to rely on the manager's decision as evidence of the original assessment. As the manager begins to revise that assessment, the minimum combined cost of wages and records can rise by more than the gain in decision value. Profit then falls locally. For an intermediate range of initial-record prices, a higher decision-record price reverses the profit comparison: the principal already finds it worthwhile to preserve the assessment

exactly. That contract retains its incentive effect as the manager's signal improves.

Better independent information can also make the principal purchase less initial effort. While the manager still follows the received assessment under high effort, the high-effort arrangement retains its equilibrium accuracy and cost. Under low effort, her independent signal improves the decisions available with zero wages and no record acquisition. If high effort is initially profitable and signal noise is sufficiently small, the optimal choice switches once to this low-effort alternative. Equilibrium accuracy falls. At the switch, high effort's additional decision value still exceeds its effort and record costs by the analyst's rent. The principal is indifferent because her cost includes this rent. Cheaper records can delay the switch by reducing the cost of inducing high effort.

A wage supporting high effort can also sustain low effort when the manager expects a less reliable assessment. Robust evaluation values each contract at its least profitable equilibrium. Guaranteeing high effort requires sufficient incentives under both expectations and can change which records the principal purchases. When recording the final decision is sufficiently expensive, review can make direct evaluation against the state the cheapest way to guarantee high effort. Compared with executing the received assessment, this change can preserve effort and equilibrium accuracy while increasing acquisition and total costs. The additional cost arises from the manager's response to a low-effort expectation, even though her high-effort decisions are unchanged.

Related literature

The contracting problem builds on the relation between information about hidden effort and the cost of incentives (Holmström, 1979). The principal also chooses which information to acquire. Li and Yang (2020) jointly choose compensation and a costly classification of performance data, and Georgiadis and Szentes (2020) design monitoring and wages to implement effort at minimum total cost. Here, the principal purchases noisy records of an assessment, the state, and a subsequent decision. The distribution of those records depends on how the manager uses the assessment. Optimal compensation and record acquisition must therefore account for the manager's response to the effort they induce.

The distinction between information for decisions and information for incentives has a substantial accounting literature. Dye (1985) shows that disclosure can make a stock price more informative about earnings while weakening its usefulness for rewarding managerial effort. Investors' responses to disclosure change the information aggregated in the price. Narayanan and Dávila (1998) study the interaction between the performance-evaluation and investment-decision uses of accounting signals. In this paper, the final decision combines information that serves different purposes: it helps the organization act correctly and provides evidence about the analyst's earlier work. Acquiring the underlying assessment and state records allows the principal to adjust compensation when better managerial evidence changes this relationship.

Managerial review also links compensation to intervention after hidden effort. Levitt and Snyder (1997) show how cancellation following an employee's early warning affects effort incentives and jointly optimize compensation, reporting, and intervention. Arya and Glover (2008) show that selective correction of erroneous performance measures can strengthen incentives because effort creates opportunities for favorable corrections. They also examine contracts using both preliminary and final scores. Here, the assessment contributes to the final decision, and the manager's anticipated reliance on it determines what agreement between the two stages reveals about effort. The optimal agreement bonus combines these effort comparisons, and the principal chooses how precisely to observe each of them.

Hidden effort in information production is central to delegated information acquisition. Chade and Kovrijnykh (2016) characterize compensation when an agent acquires information for a subsequent

investment decision. Migrow and Squintani (2023) design incentives for information acquisition and sharing across divisions; information produced in one division can make another division's performance the best basis for pay. Here, the analyst produces information that an accuracy-maximizing manager uses to make a decision. The principal chooses record precisions and wages in anticipation of that use. This permits a comparison between the gain in decision value from better managerial information and the resulting cost of compensating the analyst who supplies the initial assessment.

The manager's inference is related to observational learning. Bikhchandani et al. (1992) and Smith and Sørensen (2000) show how reliance on predecessors affects the private information revealed by actions. Information production and observation are themselves choices in Burguet and Vives (2000), Ali (2018), and Song (2016). Their analyses connect incentives to acquire information with the information available for subsequent decisions. In the present model, a manager's agreement with an assessment may reflect reliance on that assessment or support from independent evidence. These sources of agreement have different implications for rewarding the effort that generated the assessment.

Downstream actions can also be used directly in incentive schemes. Smith et al. (2021) implement efficient sequential learning with transfers that depend on an individual's action and a successor's action. In Gibbons (1988), an arbitrator combines independent evidence with information inferred from strategic offers; separating equilibria reveal the parties' private signal to the arbitrator. The analyst in this paper chooses hidden effort before the assessment generated by that effort reaches the manager. The value of acquiring the manager's decision record for payment depends on how much independent evidence the decision contains.

Organizational design changes the incentives available at each stage of production. Aghion and Tirole (1997) relate formal authority to information acquisition and effective control. Hemmer (1998) and Maskin et al. (2000) show how responsibility assignments and organizational form change the information in performance measures. In sequential production, the responses of later workers can strengthen earlier workers' incentives (Winter, 2006). Decision rights are fixed in the present model; the principal chooses compensation and record acquisition. The manager's information determines how her decision enters the analyst's incentive constraint. The principal adjusts the records and wages used to reward the initial assessment.

Inducing effort in every equilibrium can require a different contract from supporting effort in one equilibrium. Winter (2004) characterizes differential success rewards that eliminate shirking equilibria in team production, and Halac et al. (2021) use private bonuses and uncertainty about ranks to lower the cost of unique implementation. Halac et al. (2024) show that finer monitoring can be valuable for unique implementation even when it has no value for partial implementation. Here, multiplicity arises because the analyst's anticipated effort changes the manager's use of the assessment. A wage that encourages effort when the manager expects reliable work may provide weaker incentives under a different expectation. The principal can respond by changing both the bonus and the records on which it is paid.

Recent work connects implementation requirements to the choice of evaluation evidence. Camboni and Carnehl (2025) find that input tests are optimal under favorable equilibrium selection, whereas a designer concerned about adverse equilibria combines input and output tests. Competitive market payments in their model respond to certification. Ichihashi et al. (2026) study knowledge production in which agents' information evaluates their peers and derive a monitoring hierarchy that makes effort uniquely rationalizable. The principal in this paper publicly commits to a wage for one analyst and evaluates the resulting continuation equilibria. The cost of guaranteeing effort determines when the principal purchases the final-decision record and when she evaluates the initial assessment directly against the state.

Finally, Martinelli (2022) examines better voter information when political rewards are also optimized. Information affects current discipline and the selection of politicians for higher office, and voters can lose after rewards adjust. Here, the manager's independent signal changes the evidence available to compensate initial information production. Record prices determine how the principal responds to a change in the manager's use of the assessment. Profit can fall because the same effort becomes more costly to induce; equilibrium accuracy can fall because the low-effort alternative becomes more attractive.

2 Model

An analyst investigates an unknown state and produces an assessment. A manager reviews the assessment and makes the final decision. A principal rewards the analyst's work. She chooses both the wage schedule and the records used to determine payment.

2.1 Information and decisions

The state is $\theta \in \{-1, +1\}$, with equal prior probabilities. Before receiving information, the analyst chooses effort $e \in \{L, H\}$. High effort costs $\kappa > 0$; low effort is costless. Effort generates an initial assessment $a_0 \in \{-1, +1\}$ with

$$\Pr(a_0 = \theta \mid \theta, e) = \alpha_e, \qquad \tfrac{1}{2} < \alpha_L < \alpha_H < 1.$$

Thus high effort increases the probability of a correct assessment by $\Delta = \alpha_H - \alpha_L$. Effort is unobserved. The assessment is the output of this technology and is passed to the manager; the analyst chooses effort before learning what it will be.

The manager receives a noisy observation $M_0 \in \{-1, +1\}$ of the assessment, with $\Pr(M_0 = a_0 \mid a_0) = r_0 \in (1/2, 1)$. She also receives state information independently of the assessment, conditional on θ. This information takes the form

$$S = Z_r + \tau N, \qquad \Pr(Z_r = \theta \mid \theta) = r \in (1/2, 1), \qquad Z_r \in \{-1, +1\}, \quad N \sim \mathcal{N}(0, 1), \quad \tau > 0.$$

The binary signal Z_r is observed with continuous noise. The manager sees the combined signal S, together with M_0. The observation error in M_0 and the noise N are independent of all other primitive errors.

The manager chooses $d \in \{-1, +1\}$ to maximize the probability of a correct final decision. Her choice uses both the realization of her own signal and the accuracy she expects from the analyst's assessment. Neither the analyst nor the manager observes the state before this choice. The manager bears no wage or record acquisition costs.

2.2 Contracts and record acquisition

Before the analyst works, the principal publicly commits to a contract. She chooses how precisely to record three objects: the initial assessment, the state, and the final decision. After the manager decides, these measurements become available as the contractible record $R = (\widehat{a}_0, \widehat{\theta}, \widehat{d}) \in \{-1, +1\}^3$. The chosen precisions $p = (p_0, p_\theta, p_d) \in [1/2, 1]^3$ satisfy

$$\Pr(\widehat{a}_0 = a_0 \mid a_0) = p_0, \quad \Pr(\widehat{\theta} = \theta \mid \theta) = p_\theta, \quad \Pr(\widehat{d} = d \mid d) = p_d.$$

The recording errors are symmetric, mutually independent, and independent of the production and observation errors. The records determine payment; the manager makes her decision using (M_0, S).

Write $x = 2p_0 - 1$, $y = 2p_\theta - 1$, and $z = 2p_d - 1$ for the centered record precisions. A centered precision of zero gives an uninformative record, and a precision of one gives an exact record. Acquisition costs are

$$C(x, y, z) = \frac{1}{2}(c_0 x^2 + c_\theta y^2 + c_d z^2), \qquad c_0, c_\theta, c_d > 0.$$

The principal can purchase any $(x, y, z) \in [0, 1]^3$ and pays this cost in addition to wages. The prices may differ across the three records.

A contract specifies these precisions and a finite, nonnegative wage $w(R)$. The analyst is risk neutral, has outside option zero, and receives expected wage less effort cost. The principal values a correct decision at $v > 0$. Her expected profit is

$$v \Pr(d = \theta) - \mathbb{E}[w(R)] - C(x, y, z).$$

She therefore purchases records for their effect on the cost of motivating the analyst. The manager's information determines how the resulting initial work contributes to the decision.

The timing is as follows. The principal commits to records and wages. The analyst chooses effort and produces the assessment. The manager observes her signals and decides. The records are then acquired and the analyst is paid.

2.3 Equilibrium and the principal's problem

Fix a contract. Let $\sigma \in [0, 1]$ be the probability that the analyst chooses high effort. The resulting initial assessment accuracy is $q = \alpha_L + \sigma\Delta$. Write δ_q for the manager's optimal decision rule when she expects initial assessment accuracy q. A perfect Bayesian continuation equilibrium requires that this expectation agree with the analyst's strategy and that each decision maker act optimally.

For the analyst, the relevant comparison is between the expected payments from high and low effort under the same managerial rule. A secret effort deviation changes the distribution of the assessment while the manager continues to use δ_q. In particular, a high-effort equilibrium requires

$$\mathbb{E}[w(R) \mid H, \delta_{\alpha_H}] - \mathbb{E}[w(R) \mid L, \delta_{\alpha_H}] \geq \kappa.$$

At a low-effort equilibrium this difference, evaluated under δ_{α_L}, is at most κ. An interior effort probability requires equality under δ_q. These incentive conditions also ensure participation: the analyst can obtain a nonnegative payoff by choosing costless low effort.

An effort target specifies the probability σ of high effort, or equivalently the initial assessment accuracy $q = \alpha_L + \sigma\Delta$. A contract supports this target when it has an equilibrium with that probability of high effort.

The principal chooses records and wages subject to these equilibrium requirements. Write $A(q; r)$ for equilibrium accuracy: the probability that $d = \theta$ when the initial assessment has accuracy q and the manager uses δ_q. Under *favorable evaluation*, the principal evaluates a contract at its highest-profit continuation equilibrium. Equivalently, she solves

$$\begin{aligned} V^F(r) = \sup_{p,w,q} \quad & vA(q; r) - \mathbb{E}[w(R) \mid q, \delta_q] - C(x, y, z) \\ \text{subject to} \quad & p \in [1/2, 1]^3, \quad 0 \leq w(R) < \infty \text{ for every } R, \\ & q \in [\alpha_L, \alpha_H] \text{ is supported in equilibrium by } (p, w). \end{aligned}$$

The expectation at q averages over the analyst's equilibrium effort choice and all signal and recording errors. Optimal effort under favorable evaluation is the effort in a contract–equilibrium pair that

maximizes the principal's profit. A contract with zero wages and no informative records induces low effort uniquely. It gives profit $vA(\alpha_L; r)$ and is always available.

A contract can also have equilibria yielding different profits. Under *robust evaluation*, its value is the lowest profit among its continuation equilibria; $V^R(r)$ is the supremum of that value over contracts. Section 5 studies this criterion. The analysis first solves the favorable problem by finding the cheapest contract supporting each effort target and then choosing among those targets.

3 Optimal wages and record acquisition

The contract affects the accuracy the manager expects from the assessment and hence how she uses it. Her decision rule determines how the available record comparisons respond to effort and how much the principal must pay to obtain it. When she relies on the assessment, effort has a strong effect on whether the final decision is correct. When she relies on independent evidence, her decision can help check the analyst's judgment. The principal chooses payments and records to use this evidence at least cost.

3.1 Managerial decisions and effort incentives

If the initial assessment has accuracy q, the assessment received by the manager has accuracy

$$\beta(q) = r_0 q + (1 - r_0)(1 - q).$$

The first term corresponds to a correct assessment received correctly; the second corresponds to an incorrect assessment reversed by the observation error. The manager weighs M_0 using this accuracy.

A positive realization of S favors state $+1$, and a negative realization favors state -1. Based on S alone, the posterior probability of either state is at most r. Consequently, if $\beta(q) \geq r$, the manager always chooses $d = M_0$. If $\beta(q) < r$, sufficiently strong conflicting state evidence makes her override M_0. There is a threshold $s_q > 0$: she follows M_0 for $|S| < s_q$ and the sign of S for $|S| > s_q$. Ties have probability zero. A higher expected initial assessment accuracy raises the evidence required to override it. Appendix A.1 derives the threshold.

Two probability differences connect this decision rule to the analyst's incentive. Define u_q and t_q by

$$\Pr(d = \theta \mid H, \delta_q) - \Pr(d = \theta \mid L, \delta_q) = \Delta u_q,$$
$$\Pr(a_0 = d \mid H, \delta_q) - \Pr(a_0 = d \mid L, \delta_q) = \Delta t_q.$$

The coefficient u_q measures how much an improvement in initial assessment accuracy raises the probability of a correct decision, holding the manager's rule fixed. The coefficient t_q measures how much it raises the probability of agreement with the final decision under that same rule. These are consequences of the manager's Bayesian rule; they vary with her expected assessment accuracy and with her independent information.

Consider first a manager who always follows M_0. Write $m = 2r_0 - 1$. Her decision has $u_q = m$ and $t_q = 0$. Under this rule, high effort raises the probability of a correct decision, while the probability that $a_0 = d$ is r_0 under either effort. An analyst whose assessment is copied receives the same probability of agreement whether the assessment is correct or incorrect. When the manager also uses independent state evidence, t_q is positive: a more accurate assessment is more likely to agree with a decision informed by that evidence.

The principal must obtain records of the objects she wishes to compare. Independent recording errors reduce each effort-induced probability difference by the product of the relevant centered precisions:

Payment event	Increase in probability under high effort
$\widehat{a}_0 = \widehat{\theta}$	Δxy
$\widehat{a}_0 = \widehat{d}$	$\Delta t_q xz$
$\widehat{\theta} = \widehat{d}$	$\Delta u_q yz$

The first event compares the analyst's assessment directly with the state. The second uses the manager's independent evidence to check the assessment. The third uses the state record to evaluate a decision that partly reflects the analyst's work. Each individual record is equally likely to be -1 or $+1$ under either effort, so its relation to another record supplies the incentive information.

The manager's expectation matters for these comparisons. Where she uses both signals, a higher q increases u_q and reduces t_q: she relies more on the assessment and requires stronger evidence to reverse it. At fixed record precisions, a bonus for agreement between the state and final-decision records then provides a weakly greater return to effort. A bonus for agreement between the initial-assessment and final-decision records provides a weakly smaller return. This dependence allows the same contract to sustain different effort expectations. For each proposed target, the principal must therefore compute the wage using the managerial rule appropriate to that target.

3.2 Optimal wages

Fix record precisions and a positive probability of high effort, represented by initial assessment accuracy $q \in (\alpha_L, \alpha_H]$. Supporting this probability requires high effort to be optimal when $q = \alpha_H$, or indifference between efforts when q is interior. Which record realizations should be rewarded, and how much should the principal pay?

For any event E that becomes more likely under high effort, a bonus b supplies the additional expected payment $b[\Pr(E \mid H, \delta_q) - \Pr(E \mid L, \delta_q)]$. The smallest bonus that makes the analyst willing to choose high effort sets this amount equal to κ. Its expected cost at the target is therefore

$$\kappa \frac{\Pr(E \mid q, \delta_q)}{\Pr(E \mid H, \delta_q) - \Pr(E \mid L, \delta_q)}.$$

The denominator measures the incentive bought by a unit bonus; the numerator measures how often that bonus must be paid. An event that is common under both efforts can require a large wage bill even if it is more common under high effort.

There are four possible patterns among three binary records: all agree, or one of the three differs from the other two. Symmetry permits the principal to give the same payment to both sign realizations of each pattern. Comparing their expected cost per unit of incentive yields agreement among all three records as a cheapest payment event. A wage paying on several patterns combines their costs and incentives and cannot improve on the cheapest one. Appendix A.2 verifies the four comparisons.

Let $k = \kappa/\Delta$ be the cost per unit improvement in initial assessment accuracy, and put

$$K_q = xy + t_q xz + u_q yz.$$

The probability of $\widehat{a}_0 = \widehat{\theta} = \widehat{d}$ rises by $\Delta K_q/2$ under high effort. Thus a bonus of $2k/K_q$ supplies exactly the additional expected wage κ. To calculate its expected cost, write $h_q = 2q - 1$ and $N_q = 1 + u_q xz + t_q yz + h_q K_q$. The probability of agreement among all three records at the target is $N_q/4$.

Proposition 1. *Fix $q \in (\alpha_L, \alpha_H]$ and record precisions $(x, y, z) \in [0, 1]^3$. If $K_q > 0$, the wage*

$$w_q(R) = \frac{2k}{K_q} \mathbf{1}\{\widehat{a}_0 = \widehat{\theta} = \widehat{d}\}$$

minimizes expected payment among all finite nonnegative wages supporting q under the manager's rule δ_q. The minimum is

$$W_S(q; x, y, z) = \frac{kN_q}{2K_q} = \frac{k}{2}\left(2q - 1 + \frac{1 + u_q xz + t_q yz}{xy + t_q xz + u_q yz}\right). \tag{1}$$

If $K_q = 0$, no wage on these records supports q.

A more precise record can reduce the required wage by making the rewarded event more responsive to effort relative to its frequency.

To see what this wage reduction saves, consider the analyst's payoff under the optimal bonus. Its incentive constraint binds, so choosing high or low effort gives the same net payoff under the target rule δ_q. The analyst's equilibrium rent is therefore the expected wage she would receive from choosing costless low effort under that rule:

$$W_S(q; x, y, z) - \kappa\sigma = \mathbb{E}[w_q(R) \mid L, \delta_q], \qquad \sigma = \frac{q - \alpha_L}{\Delta}.$$

Even low effort sometimes produces a correct assessment. With exact assessment and state records, paying k whenever the assessment is correct supplies the required incentive and leaves rent $k\alpha_L$. With noisy evidence, an incorrect assessment may also earn the bonus. Maintaining the required payment advantage for correct work then increases the wage bill. High effort raises the probability of a correct assessment by Δ, so the binding incentive constraint makes the conditional payment for a correct assessment exceed that for an incorrect assessment by k. The rent decomposition is

$$W_S(q; x, y, z) - \kappa\sigma = k\alpha_L + \mathbb{E}[w_q(R) \mid a_0 \neq \theta, \delta_q].$$

More precise records can reduce payments made when the initial assessment is wrong. When wages adjust to sustain the same effort probability, expected effort cost remains unchanged, so the wage saving reduces the analyst's rent by the same amount. With the manager's signal precision also unchanged, the manager uses the same decision rule and equilibrium accuracy is unchanged. The principal purchases more precise evidence when the reduction in expected wages covers the additional acquisition cost. She can therefore spend more on records and less in total to obtain the same effort and equilibrium accuracy.

3.3 Optimal record acquisition

The principal balances the wage saving from more precise evidence against its acquisition cost. For a target $q \in (\alpha_L, \alpha_H]$, her minimum total cost is

$$T_S(q; r) = \min_{(x,y,z)\in[0,1]^3} \{W_S(q; x, y, z) + C(x, y, z)\}. \tag{2}$$

A zero incentive denominator is assigned infinite cost. The minimum is attained. Any minimizing precisions, together with the bonus in Proposition 1, give a least-cost contract supporting the target. At an interior precision choice, the marginal expected wage saving equals the marginal acquisition cost. A record is exact when its wage saving justifies paying for precision up to one.

The principal can evaluate the assessment using only its record and the state record. With $z = 0$, averaging out the uninformative decision record in the optimal wage yields

$$w_D(R) = \frac{k}{xy}\mathbf{1}\{\widehat{a}_0 = \widehat{\theta}\}.$$

High effort raises this event's probability by Δxy, so the bonus supplies exactly κ in additional expected payment. The expected wage is $k[2q-1+(xy)^{-1}]/2$. Choosing the two precisions therefore amounts to

$$\min_{0<x,y\leq 1}\left\{\frac{k}{2xy} + \frac{c_0x^2 + c_\theta y^2}{2}\right\}. \tag{3}$$

When both choices are interior, the first-order conditions give $c_0x^2 = c_\theta y^2 = k/(2xy)$. Equal expenditure on the two records balances their contributions to the comparison, and the cheaper record is acquired more precisely. Denote the unique minimizing precisions by (x_D, y_D), and their acquisition cost by $C_D = (c_0x_D^2 + c_\theta y_D^2)/2$. Direct evaluation costs

$$T_D(q) = \frac{k}{2}\left(2q - 1 + \frac{1}{x_D y_D}\right) + C_D.$$

The precision choices are the same for every positive probability of high effort. The bonus has the same effect on effort under every managerial rule because its payment event concerns only the assessment and the state. Appendix A.3 gives the solution, including exact recording.

Acquiring a final-decision record can lower this cost by supplying evidence that would otherwise require more precise assessment or state records. When the manager relies on the assessment, her decision provides a second observation of it. When she uses independent state information, her decision can help check whether the assessment was correct. In the wage formula, these two uses correspond to u_q and t_q.

At the direct optimum $(x_D, y_D, 0)$, the marginal wage saving from recording the decision is proportional to

$$u_q y_D(1 - x_D^2) + t_q x_D(1 - y_D^2).$$

The first term is positive when the manager uses the assessment and its direct record is imperfect. The second is positive when she uses independent evidence and the state record is imperfect. Whenever this sum is positive, a sufficiently small positive centered precision z saves wages in proportion to z, at an acquisition cost proportional to z^2. Some decision recording is then optimal at every finite price c_d. At high prices the principal buys little of it.

If the direct records already supply all of this evidence, the principal can stop recording the decision at a finite price. For a high-effort target, define $c_S^*(r)$ as the supremum of prices c_d for which including the decision strictly lowers total cost below $T_D(\alpha_H)$; set it to zero if there are no such prices. Thus $T_S(\alpha_H; r) < T_D(\alpha_H)$ exactly for $0 < c_d < c_S^*(r)$. Above the threshold all optimal record choices have $z = 0$. The threshold may be infinite. If the manager expecting high effort always follows M_0, it is finite exactly when the direct alternative records the assessment exactly. If she also uses independent evidence, both direct records must be exact for the threshold to be finite. Appendix A.5 gives the full characterization.

The manager's use of the assessment affects both the value of producing it and the cost of rewarding its production. The principal chooses the effort probability by comparing its decision value with the optimized wage and record acquisition cost $T_S(q; r)$. Write $A_L(r) = A(\alpha_L; r)$ and $A_H(r) = A(\alpha_H; r)$. Her favorable value is

$$V^F(r) = \max\left\{vA_L(r), \sup_{q\in(\alpha_L,\alpha_H]}[vA(q; r) - T_S(q; r)]\right\}.$$

The zero-wage alternative supplies the first term. A target $q > \alpha_L$ improves decision value by $v[A(q;r) - A_L(r)]$ and requires total cost $T_S(q;r)$. It is worth purchasing when the improvement exceeds that cost.

4 Managerial information and the optimal contract

An increase in r makes the manager's independent signal $S = Z_r + \tau N$ more informative in the Blackwell order. The effort technology, observation precision r_0, noise τ, decision value v, and record prices remain fixed. The principal can change wages, all three record precisions, and the effort she purchases.

At a fixed initial assessment accuracy, the manager benefits from the better signal. Equilibrium accuracy weakly increases, and increases strictly once $r > \beta(q)$ and she sometimes overrides M_0. Her decision then relies less on the assessment and more on independent evidence: u_q decreases and t_q increases. Keeping the same effort after this change may require a different bonus or different record precisions. The principal's gain from better decisions must therefore be assessed together with the cost of sustaining the initial work on which those decisions rely.

4.1 Record precision as managerial information improves

Consider contracts that sustain the same probability of high effort as the manager's independent information improves. Fix the corresponding initial assessment accuracy $q \in (\alpha_L, \alpha_H]$ and compare $r_1 < r_2 \leq \bar{r} < 1$, with $r_2 > \beta(q)$ so that the manager's rule changes. At each r_i, the principal chooses wages and records to support q at minimum total cost. Denote any minimizing record choices by (x_i, y_i, z_i).

When the state-record price is sufficiently low, the principal acquires an exact state record. In particular, if $c_\theta < k(1-\bar{r})/(2\bar{r})$, every optimal record choice has $y = 1$ for every positive probability of high effort and every $r \leq \bar{r}$. With the state known, the principal still needs evidence of what the analyst originally concluded. She can obtain this evidence from the initial record or from a decision that reflects the assessment.

As the manager relies more on independent information, her decision reveals less about the original assessment. The principal responds by preserving that assessment more precisely:

$$x_2 \geq x_1.$$

This comparison allows the bonus and final-decision record precision to adjust. At unchanged record precisions with $y = 1$, the expected wage required to support q weakly rises, strictly so when $x < 1$ and $z > 0$. Even after optimizing the records, the minimum total cost of supporting the target weakly rises over this information comparison. The principal must replace evidence previously supplied by the decision or pay a larger wage for the remaining evidence.

If also $c_0 \leq 2k$, final-decision record precision weakly falls: $z_2 \leq z_1$. The organization acquires a clearer initial record and relies less on recording the manager's decision. The condition on c_0 ensures that replacing the lost evidence through the initial record is inexpensive enough to yield this direction of adjustment. When initial recording is sufficiently costly, final-decision record precision can rise: the principal purchases a more precise observation of a decision that has become less informative about the initial assessment.

A sufficiently low price for recording the initial assessment makes that record exact. If

$$c_0 \leq \frac{k(1-m)}{2(1+m)}, \qquad m = 2r_0 - 1, \tag{4}$$

every optimal record choice has $x = 1$, for every positive probability of high effort and every $r \in (1/2, 1)$. Given the analyst's original judgment, the principal can use the decision to help determine whether that judgment was correct. As the manager's independent evidence improves, the decision supplies more of the state information needed for this comparison. The principal weakly reduces state-record precision:

$$y_2 \leq y_1.$$

At unchanged record precisions with $x = 1$, the minimum expected wage weakly falls as r rises. Optimizing the records can reduce cost further. If also $c_\theta \leq 2k$, final-decision record precision weakly rises, $z_2 \geq z_1$: the organization uses a more precise decision record while spending less on the state record.

The two comparisons concern different uses of the manager's decision. With the state known, the principal needs to establish what the analyst originally concluded. More independent managerial evidence makes the decision less useful for that purpose. With the initial assessment known, the principal needs to establish whether it was correct. More independent evidence makes the decision more useful for this comparison. Record prices determine which of these uses matters for compensation.

When $r_1, r_2 \leq \beta(q)$, the manager always follows M_0 and the set of optimal record choices is unchanged. Appendices B.2 and B.3 prove the comparisons, including the boundaries at zero and exact recording.

When initial recording costs satisfy (4), supporting cost $T_S(q; r)$ weakly falls at every target while equilibrium accuracy weakly rises. Every previous target becomes at least as profitable, so $V^F(r)$ is nondecreasing. When the state-record price is sufficiently low, supporting cost moves in the opposite direction. Profit then depends on whether the additional decision value covers the increase in minimum total cost.

4.2 Profit when high effort remains optimal

Suppose the state record is inexpensive enough to acquire exactly. A decision that follows the analyst's assessment can help the principal economize on recording the assessment itself. Once the manager begins to revise it, the principal loses some of this evidence. A high value of accurate decisions can still make high effort worth purchasing, even though it has become more expensive to obtain.

The price of the final-decision record determines whether the organization was using this source of evidence. Let $\beta_H = \beta(\alpha_H)$, the accuracy of the assessment received by the manager after high effort. At $r = \beta_H$ the manager always follows M_0 under high effort. A sufficiently inexpensive decision record can then make an imperfect initial record optimal. At prices above $c_S^*(\beta_H)$, direct evaluation already attains the lowest cost.

For sufficiently low state-record prices, the principal verifies the state exactly as r increases. For sufficiently high decision values, reducing effort sacrifices more decision value than it saves in wages and records. For an upper signal precision $\bar{r} > \beta_H$, the restrictions below keep $y = 1$ and effort high throughout $r \leq \bar{r}$. Write $u_L(r) = u_{\alpha_L}(r)$ and recall that $T_D(\alpha_H)$ is the optimized cost of obtaining high effort using assessment and state records alone.

Proposition 2. *Fix $\bar{r} \in (\beta_H, 1)$ and suppose*

$$c_\theta < \frac{k(1-\bar{r})}{2\bar{r}}, \qquad v > \max\left\{\frac{k}{u_L(\bar{r})}, \frac{T_D(\alpha_H)}{A_H(\bar{r}) - A_L(\bar{r})}\right\}.$$

For every $c_d > 0$, high effort is the unique optimal effort under favorable evaluation throughout $r \in (1/2, \bar{r}]$. Every minimum-cost record choice supporting high effort has $y = 1$.

1. *If* $c_d < c_S^*(\beta_H)$*, every optimal record choice at* β_H *has* $0 < x < 1$ *and* $z > 0$*, and* $V^F(r) < V^F(\beta_H)$ *for all sufficiently small increases in* r *above* β_H*.*

2. *If* $c_d \geq c_S^*(\beta_H)$*, direct evaluation minimizes cost at* β_H*, and* $V^F(r) > V^F(\beta_H)$ *for every* $r \in (\beta_H, \bar{r}]$*.*

In both cases, equilibrium accuracy is $A_H(r) > \beta_H$ *for* $r \in (\beta_H, \bar{r}]$*.*

Just above β_H, the manager first revises the assessment in cases where the two actions have almost the same posterior probability of being correct. Those revisions produce a small increase in equilibrium accuracy relative to their effect on the relation between the assessment and the final decision. With an exact state record and an imperfect initial record, this relation was helping the principal reward the analyst. Its weakening raises the minimum combined cost of wages and records by more than the additional decision value.

The organization responds by adjusting its bonus and record precisions. Continuing to use the decision requires adjusting the bonus or other record precisions as the decision reveals less about the assessment. Switching to direct evaluation requires paying for evidence previously obtained more cheaply through the decision. Near the point where revisions begin, the increase in minimum total cost exceeds the gain in decision value. High effort remains optimal: reducing effort would sacrifice more decision value than it saves in wages and record acquisition.

If direct evaluation was already optimal, the principal can keep its wage and records as r rises. The analyst then has the same incentive, and the manager makes more accurate decisions. Profit therefore exceeds its value at $r = \beta_H$ for every $r \in (\beta_H, \bar{r}]$.

The threshold $c_S^*(\beta_H)$ is finite and strictly positive when

$$\frac{k(1-m)}{2(1+m)} < c_0 < \frac{k}{2}. \tag{5}$$

With these initial-record prices, changing c_d alone can reverse the profit effect of a small information improvement. When recording the decision is cheap, the principal uses it to economize on preserving the original assessment. The resulting contract depends on the manager continuing to convey information about that assessment. Better independent evidence weakens this source of incentives. When recording the decision is expensive, the principal already preserves the assessment exactly and evaluates it against the state. That contract continues to provide the same incentive as the manager's information improves. The different profit responses follow from the records the principal found worth purchasing before the improvement.

For the prices and decision value in Proposition 2, if the initial-record price also satisfies (5), an increase from $r = \beta_H$ to $r \in (\beta_H, \bar{r}]$ leads the principal to weakly increase initial-record precision and weakly reduce final-decision record precision. Effort remains high. When $c_d < c_S^*(\beta_H)$, the optimized sum of wages and record costs exceeds its value at $r = \beta_H$ for all sufficiently small increases in r. If initial recording is more expensive, $c_0 > k/2$, then $c_S^*(\beta_H) = \infty$: some decision recording remains optimal at every finite c_d, and the local profit decline occurs at every such price. Appendix B.4 proves these comparisons.

4.3 The choice between high and low effort

The principal can also respond to better managerial information by buying less initial effort. The relevant benefit of high effort is its decision value relative to the outcome available with zero wages. Independent managerial information improves this alternative and can reduce the amount the principal is willing to spend on initial work. Within this signal family, increasing r weakly reduces

the equilibrium accuracy gain from switching from low to high effort. At each expected initial assessment accuracy, the manager relies less on the assessment as her independent evidence improves. Initial effort and independent evidence are substitutes in producing accurate decisions.

In a benchmark without review, the organization executes the received assessment, $d = M_0$. The Bayesian manager makes the same decision under every effort expectation when $r \leq \beta_L$, where $\beta_L = \beta(\alpha_L)$. The observation error between the analyst and the decision is the same in both arrangements. In this benchmark, equilibrium accuracy is $\beta(q)$ and the expected wage formula has $(u_q, t_q) = (m, 0)$. Let T_0 denote the minimum total cost of obtaining high effort in this benchmark, after all wages and records are optimized. The corresponding cost for another target $q > \alpha_L$ is $T_0 + k(q - \alpha_H)$. Both decision value and supporting cost are affine in the target. Comparing all effort probabilities therefore gives

$$V_0 = \max\{v\beta_L,\ v\beta_H - T_0\}. \tag{6}$$

The organization either purchases high effort with the least-cost agreement contract or pays zero and accepts low effort. Appendix A.6 gives the cost formula and the contract when the initial assessment and state records are inexpensive.

As r rises above β_L, the manager begins to revise assessments when she expects low effort. Under high effort she continues to follow M_0 until r exceeds β_H. Throughout $\beta_L < r \leq \beta_H$, the high-effort arrangement consequently retains equilibrium accuracy β_H and cost T_0, while the zero-wage arrangement becomes more accurate. If noise is small enough, every interior target yields less profit than either high effort or zero wages. Write Φ for the standard normal distribution function.

Proposition 3. *Suppose*

$$v(\beta_H - \beta_L) > T_0, \qquad v\Phi(-1/\tau) < k\alpha_L.$$

There is a unique $r^* \in (\beta_L, \beta_H)$ satisfying

$$v[\beta_H - A_L(r^*)] = T_0. \tag{7}$$

Under favorable evaluation, the unique optimal effort is H for $r < r^*$ and L for $r > r^*$. Both pure efforts are optimal at r^*. For $r \in (1/2, 1)$, optimized profit is

$$V^F(r) = \begin{cases} v\beta_H - T_0, & r \leq r^*, \\ vA_L(r), & r \geq r^*. \end{cases}$$

Profit is continuous and nondecreasing. Switching from H to L at r^* lowers equilibrium accuracy by T_0/v.

The inequality $v(\beta_H - \beta_L) > T_0$ makes high effort worth buying when the received assessment is executed. The bound on $\Phi(-1/\tau)$ limits the equilibrium accuracy gain from an intermediate target where the manager uses independent evidence. Such a target still requires the analyst to be willing to incur the full high-effort cost whenever she chooses H, and therefore requires positive rent even if she rarely chooses it. Under the noise bound, its decision gain is too small to cover the required wage. Appendix B.5 establishes this comparison across all intermediate targets.

For $r < r^*$, the principal can retain the benchmark's least-cost high-effort contract. For $r > r^*$, she pays zero and acquires no records. At r^* the wage and record savings from abandoning high effort exactly offset the lost decision value. Profit is continuous although effort and equilibrium

accuracy change discretely. The manager's rule remains optimal given the initial assessment's reliability. Equilibrium accuracy falls because the organization stops paying to produce the more accurate assessment.

This choice can discard work whose decision contribution still exceeds its resource cost. Write $T_0 = W_0 + C_0$, where W_0 is the expected wage and C_0 is record acquisition cost in a least-cost high-effort contract. At the threshold,

$$[v\beta_H - \kappa - C_0] - vA_L(r^*) = W_0 - \kappa \geq k\alpha_L > 0.$$

High effort generates greater decision value net of effort and record costs. The principal is nevertheless indifferent because she must also pay the analyst's rent. Abandoning effort saves her the entire wage, while it saves effort resources only by κ.

Record prices affect the threshold through the optimized cost T_0. A reduction in prices weakly lowers T_0, delays the exit to a higher r^*, and reduces the equilibrium accuracy loss T_0/v, provided both price vectors satisfy the proposition's conditions. These effects are strict when the price reduction lowers T_0. Less expensive evidence lets the organization retain high effort over a wider range of managerial information.

5 Contracts that guarantee high effort

The minimum wage in Proposition 1 makes the analyst willing to supply the target effort when the manager expects it. The same wage can give a weaker incentive when she expects poorer work and relies more on her own evidence. A principal evaluating a contract at its lowest-profit equilibrium must take this response into account. She may need to change what the wage rewards, how precisely it is recorded, or the size of the bonus.

5.1 The cost of guaranteeing high effort

Consider an optimal supporting contract with an exact state record and an informative final-decision record, $y = 1$ and $z > 0$. These are the record choices in the local profit-loss comparison of Proposition 2. Write $K_L = K_{\alpha_L}$ and $K_H = K_{\alpha_H}$, and use the same subscripts for u_q, t_q, N_q. For $r > \beta_L$, the bonus event is more responsive to effort when the manager expects high effort: $K_H > K_L$. Thus the bonus $2k/K_H$, which just supports high effort, gives an analyst facing a low-effort expectation net gain

$$\frac{2k}{K_H}\frac{\Delta K_L}{2} - \kappa = \kappa\left(\frac{K_L}{K_H} - 1\right) < 0$$

from choosing high effort. Low effort is also an equilibrium.

The reason is the manager's dependence on the assessment. With an exact state record, the contract uses the decision partly as evidence of what the analyst initially concluded. A manager expecting more reliable work uses it more. The same bonus then provides a larger return to producing an accurate assessment. Formally, where the manager uses both signals, $u'_q > 0$ and $t'_q = -(2q-1)u'_q$, giving $K'_q = zu'_q[y - (2q-1)x]$. This derivative is positive when $y = 1$ and $z > 0$. This dependence of incentives on the manager's expectation allows both the high- and low-effort expectations to be self-consistent.

To guarantee high effort, the principal must provide a sufficient incentive at every possible effort expectation. For an agreement bonus, the weakest incentive occurs at one of the two endpoints:

$$\min_{q\in[\alpha_L,\alpha_H]} K_q = \min\{K_L, K_H\}. \tag{8}$$

Consequently, a bonus strictly above $2k/\min\{K_L, K_H\}$ makes high effort strictly optimal under every managerial rule. The principal can reduce this required bonus by changing the records as well.

Let $T_G(r)$ be the infimum combined wage and acquisition cost of contracts that induce high effort in every equilibrium. Optimization over all nonnegative wages and record precisions gives

$$T_G(r) = \min_{(x,y,z)\in[0,1]^3} \left\{ \frac{kN_H}{2\min\{K_L, K_H\}} + C(x,y,z) \right\}. \tag{9}$$

A zero denominator is assigned infinite cost. The numerator measures the probability of payment under high effort. The denominator now uses the smaller incentive response across the two expectations. At a minimizing precision vector, multiplying the bonus $2k/\min\{K_L, K_H\}$ by $1+\varepsilon$ guarantees high effort for every $\varepsilon > 0$. Its cost converges to T_G as ε tends to zero. The strict incentive is preserved along this sequence as its cost approaches T_G. Appendix C.1 proves the lower bound over all wage schedules and the endpoint comparison.

The principal can always use the direct wage based on the assessment and state. Its effort incentive is independent of the manager's expectation. The three relevant costs therefore satisfy

$$T_S(\alpha_H; r) \le T_G(r) \le T_D(\alpha_H).$$

The infimum cost of guaranteeing high effort equals the minimum cost of supporting it exactly when some minimum-cost record choice for supporting high effort satisfies $K_L \ge K_H$. Otherwise, the principal must spend more to make effort worthwhile when the manager expects low effort. Direct evaluation bounds this extra expense.

5.2 Record choice for guaranteed high effort

The guarantee requirement changes the prices at which recording the final decision is worthwhile. Define $c_G^*(r)$ as the supremum of prices c_d for which $T_G(r) < T_D(\alpha_H)$, taking zero if there are none. The cost ordering implies $c_G^*(r) \le c_S^*(r)$. Above $c_G^*(r)$, every minimizing record choice in (9) has $z = 0$. That record must improve incentives sufficiently under both expectations to justify its acquisition cost.

Start with $r \le \beta_L$, when the manager follows M_0 under every effort expectation, and increase r to (β_L, β_H). At high effort she still follows M_0. At low effort she sometimes revises it. The minimum cost of supporting high effort remains T_0, but the incentive supplied at a low-effort expectation changes. Its difference from the high-effort incentive is

$$K_L - K_H = z\{t_L x - (m - u_L)y\}.$$

The term $t_L x$ reflects the independent evidence now supplied by the decision when compared with the initial record. The term $(m - u_L)y$ reflects the loss of the assessment's influence on the probability of a correct decision. A bonus for agreement between the state and final-decision records can therefore become more expensive when it must guarantee high effort.

This can make the organization replace the final-decision record with more expensive direct records. Consider prices for which the optimal direct records (x_D, y_D) are both imperfect. In the benchmark without review, the final decision supplements the initial record, so some decision recording reduces cost at every finite c_d. For the guarantee after review is introduced, the effect depends on how recording changes payments under high effort and incentives under a low-effort expectation. At prices satisfying the strict inequality in Lemma C.3, increasing final-decision record precision raises the frequency of bonus payments under high effort proportionally more than it improves incentives under the low-effort expectation. The expected wage needed to guarantee high

effort therefore rises at $(x_D, y_D, 0)$. Under this condition, $c_G^*(r)$ is finite. At every price above it, $(x_D, y_D, 0)$ minimizes (9), so the infimum cost of guaranteeing high effort is $T_D(\alpha_H) > T_0$.

When $v(\beta_H - r) > T_D(\alpha_H)$, high effort is worth purchasing under both evaluation criteria, including comparison with all interior targets. The favorable contract keeps its old high-effort outcome and cost. The robust value is approached by slightly increased direct bonuses that guarantee the same effort. The values are

$$V^F(r) = V_0 = v\beta_H - T_0, \qquad V^R(r) = v\beta_H - T_D(\alpha_H).$$

Effort and equilibrium accuracy are the same as in the benchmark. Under high effort, the manager continues to choose $d = M_0$. The direct bonus also makes high effort worthwhile when she expects low effort and sometimes revises the received assessment. Its incentive effect is the same under either expectation because payment depends on the initial assessment and the state. The principal pays more to acquire these records: acquisition cost rises from C_0 to C_D, and the robust profit value falls by $T_D(\alpha_H) - T_0$. The additional expense secures high effort against the competing low-effort equilibrium.

The expected wage component need not rise with total cost. In this comparison, if $x_D^2 > 1/3$, the more precise direct records reduce the expected wage at all sufficiently large finite decision-record prices. The increase in record acquisition cost more than offsets that wage saving. If $x_D^2 < 1/3$, the expected wage is higher at sufficiently large prices as well. For each such price, sufficiently small bonus increases preserve these wage comparisons. Appendix C.5 gives these component comparisons.

5.3 Effort choice under robust evaluation

The principal may prefer to reduce effort when guaranteeing high effort becomes expensive. Her comparison includes the zero-wage contract, contracts guaranteeing high effort, and contracts uniquely inducing an interior effort probability. Appendix C derives the infimum total cost for every interior target and the resulting robust value. It accounts for the small contract adjustments needed when an incentive constraint binds at a competing equilibrium.

Under the small-noise condition $v\Phi(-1/\tau) < k\alpha_L$, the same wage lower bound used in Proposition 3 excludes profitable interior effort. The robust problem then becomes

$$V^R(r) = \max\{vA_L(r),\ vA_H(r) - T_G(r)\}.$$

On $\beta_L < r < \beta_H$, favorable evaluation purchases high effort when its additional decision value $v[\beta_H - A_L(r)]$ exceeds T_0. Robust evaluation requires it to exceed $T_G(r)$. When the value lies strictly between these costs, favorable evaluation retains high effort and robust evaluation chooses low effort.

If the benchmark purchased high effort, this robust switch lowers both equilibrium accuracy and profit relative to that benchmark. Equilibrium accuracy falls from β_H to $A_L(r)$, and the profit change is $T_0 - v[\beta_H - A_L(r)] < 0$. The organization gives up initial effort because sustaining it under all equilibrium expectations has become too costly. Once the additional decision value also falls below T_0, both criteria choose low effort; the savings in wages and record acquisition cost then raise profit relative to the benchmark.

Direct evaluation limits how costly managerial review can be. It can always deliver high effort at total cost arbitrarily close to $T_D(\alpha_H)$, while review weakly improves equilibrium accuracy at that effort. If the direct contract already minimizes cost in the benchmark without review, review therefore weakly raises the robust value. More generally, the potential loss is bounded by the benchmark's saving from using the final decision, less the decision gain from review. Appendix C.4

states this bound. The comparison holds the review arrangement fixed; an organization free to choose whether to introduce review can retain the benchmark without review.

6 Conclusion

How should an organization reward initial work that another decision maker uses and revises? With the symmetric signals and independent record errors studied here, a bonus for agreement among the assessment, state, and decision records minimizes the expected wage sustaining a given positive probability of high effort in equilibrium. The principal purchases record precision to reduce the rent required to obtain that effort. The manager's decision can save record acquisition cost by supplying evidence about the original assessment or its accuracy. The records worth purchasing therefore depend on how the manager uses the assessment.

A more informative independent signal changes both the contribution of initial effort to equilibrium accuracy and the cost of purchasing it. Holding the probability of high effort fixed, a sufficiently low state-record price keeps the state record exact and makes optimal initial-record precision weakly rise as the manager changes her decisions. When initial assessments are sufficiently inexpensive to record exactly, optimal state-record precision weakly falls while sustaining the same effort probability. Supporting costs then weakly fall at every effort probability, so optimized favorable profit weakly increases even when effort is reoptimized. With a sufficiently low state-record price and an imperfect initial record, the adjustment in records can instead leave high effort more expensive to obtain. Near the onset of managerial revisions, this cost increase can reduce profit while effort stays high and decisions become more accurate.

The principal can also reduce the effort she purchases. In the region where high-effort decisions remain unchanged, a more informative independent signal improves the low-effort alternative until the principal switches to zero wages and no record acquisition. Equilibrium accuracy falls. At the switch, high effort still yields greater decision value net of effort and acquisition costs. The wage required to obtain it includes the analyst's rent. Lower record costs can delay this switch and reduce the associated loss of equilibrium accuracy.

Guaranteeing high effort requires the contract to provide incentives when the manager expects poor initial work as well as when she expects good work. This requirement can make the principal use only assessment and state records. Compared with directly executing the received assessment, the organization maintains high effort and the same equilibrium accuracy at greater record acquisition cost and total cost. The cost of rewarding initial work depends on how the manager would use it under different effort expectations. Evaluating an improvement in managerial information therefore requires accounting for the resulting changes in wages, record acquisition, and the effort the organization chooses to induce.

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A Wages and record acquisition

A.1 The manager's rule

Conditional on $\theta = +1$, the density of S is a mixture of normal densities centered at $+1$ and -1, with weights r and $1 - r$. Conditional on $\theta = -1$, the weights are reversed. The ratio of the two normal component densities at S is $\exp(2S/\tau^2)$. Conditional independence and the equal prior give posterior log odds

$$M_0 \log \frac{\beta(q)}{1-\beta(q)} + \log \frac{r\exp(2S/\tau^2) + 1 - r}{(1-r)\exp(2S/\tau^2) + r}.$$

The manager chooses $+1$ when this expression is positive and -1 when it is negative. The state-evidence likelihood ratio lies between $(1-r)/r$ and $r/(1-r)$. If $\beta(q) \geq r$, the posterior therefore always favors M_0. If $\beta(q) < r$, setting the log odds to zero at $M_0 = -1$ gives $\exp(2s_q/\tau^2) = [r + \beta(q) - 1]/[r - \beta(q)]$, which yields

$$s_q = \frac{\tau^2}{2} \log \frac{r + \beta(q) - 1}{r - \beta(q)}. \tag{10}$$

Symmetry gives the threshold $-s_q$ when $M_0 = +1$.

Symmetry permits the conditional representation

$$\mathbb{E}[d \mid a_0, \theta; \delta_q] = u_q a_0 + t_q \theta.$$

Changing actual effort from L to H changes the distribution of a_0 while holding this rule fixed. It raises the probability of a correct decision by Δu_q and agreement $a_0 = d$ by Δt_q, so these coefficients coincide with the probability-difference definitions in the text. With $h_q = 2q - 1$, equilibrium accuracy is

$$A(q; r) = \frac{1 + h_q u_q + t_q}{2}.$$

The coefficients satisfy $u_q > 0$, $t_q \geq 0$, and $u_q + t_q < 1$. For any fixed rule, the probability of a correct decision is affine in q. Maximizing over rules makes $A(q; r)$ convex, and the envelope condition gives $\partial A(q; r)/\partial q = u_q$.

Let Φ be the standard normal distribution function. With $s_q = \infty$ when $\beta(q) \geq r$, the decision coefficients are

$$\begin{aligned} u_q &= m\left[\Phi((s_q - 1)/\tau) - \Phi((-s_q - 1)/\tau)\right], \\ t_q &= (2r - 1)\left[\Phi((1 - s_q)/\tau) - \Phi((-1 - s_q)/\tau)\right]. \end{aligned}$$

At $s_q = \infty$, these formulas give $(u_q, t_q) = (m, 0)$. In the range $r > \beta(q)$, differentiating the cutoff and normal probabilities gives $u_q' > 0$ and $t_q' = -h_q u_q'$. The latter identity also follows from the envelope condition for equilibrium accuracy. It implies $dt/du = -h_q$, which decreases with q. Thus t is strictly concave as a function of u in that region. The range where $d = M_0$ contributes a single endpoint to this relation.

For $\beta(q) \leq r$, the bound $A(q; r) \leq r$ can also be seen from the binary experiment (M_0, Z_r). If the manager observes both binary signals, she follows the more accurate Z_r when they disagree, and is correct with probability r. Observing S in place of Z_r is a garbling of this experiment.

A.2 Optimal supporting wages

For a general wage, define its conditional payment difference by

$$D_w(q) = \mathbb{E}[w(R) \mid a_0 = \theta; \delta_q] - \mathbb{E}[w(R) \mid a_0 \neq \theta; \delta_q].$$

The gain from choosing high effort is $\Delta D_w(q) - \kappa$. Consequently, H is an equilibrium when $D_w(\alpha_H) \geq k$, L is an equilibrium when $D_w(\alpha_L) \leq k$, and an interior equilibrium requires $D_w(q) = k$.

Proof of Proposition 1. Simultaneously reversing the state, assessment, and decision leaves all probabilities unchanged. Averaging a wage with its sign reversal preserves its expected payment and effort incentives. Index the four resulting record classes by $s = \widehat{a}_0\widehat{\theta}$ and $j = \widehat{a}_0\widehat{d}$, each in $\{-1, +1\}$. At the fixed manager rule, define

$$L_{sj} = 1 + u_q xzj + t_q yzsj, \qquad K_{sj} = xys + t_q xzj + u_q yzsj.$$

The class probability at target q is $(L_{sj} + h_q K_{sj})/4$ and its probability difference between actual efforts is $\Delta K_{sj}/2$. Agreement among all three records corresponds to class $(s, j) = (+1, +1)$.

For classes with $K_{sj} > 0$, minimizing expected wage per unit of incentive is equivalent to maximizing K_{sj}/L_{sj}. Cross multiplication between class $(+1, +1)$ and each other class gives, writing $u = u_q$ and $t = t_q$,

$$\begin{aligned} 2z[tx(1-y^2) + uy(1-x^2)] &\geq 0, \\ 2y[x(1-t^2z^2) + u^2xz^2 + uz(1+x^2)] &\geq 0, \\ 2x[y(1-u^2z^2) + t^2yz^2 + tz(1+y^2)] &\geq 0. \end{aligned}$$

The inequalities use $x, y, z \in [0, 1]$ and $u, t \geq 0$, $u + t < 1$. Classes with $K_{sj} \leq 0$ cannot improve the minimum wage. Paying only when all three records agree, and setting its bonus to make the effort constraint bind, therefore yields $b = 2k/K_q$. Multiplication by its probability $N_q/4$ gives (1). If $K_q = 0$, the effort-induced probability difference is zero for every class, so high effort cannot be optimal. □

A.3 Optimal precision for direct evaluation

For $c_0 \leq c_\theta$, the unique minimizer is

$$(x_D, y_D) = \begin{cases} ((k/2)^{1/4} c_\theta^{1/8} c_0^{-3/8}, (k/2)^{1/4} c_0^{1/8} c_\theta^{-3/8}), & k \leq 2c_0^{3/2}/\sqrt{c_\theta}, \\ (1, (k/(2c_\theta))^{1/3}), & 2c_0^{3/2}/\sqrt{c_\theta} < k < 2c_\theta, \\ (1, 1), & k \geq 2c_\theta. \end{cases} \tag{11}$$

For $c_0 > c_\theta$, interchange the coordinates and their prices.

Derivation of the direct-record optimum. The objective in (3) is strictly convex on its positive domain. Its interior first-order conditions equate $c_0 x^2 = c_\theta y^2$ to $k/(2xy)$. Solving them gives the first row of (11). If $c_0 \leq c_\theta$, the unconstrained initial record reaches one first, at $k = 2c_0^{3/2}/\sqrt{c_\theta}$. Conditional on $x = 1$, minimization gives $y = (k/(2c_\theta))^{1/3}$ until $k = 2c_\theta$. Thereafter both coordinates equal one. The expressions agree at the boundaries. The target-dependent term $kh_q/2$ is additive, so these precisions apply to every target $q \in (\alpha_L, \alpha_H]$. □

For equal prices $c_0 = c_\theta = c > k/2$, the optimal bonus is $\sqrt{2kc}$, and

$$x_D = y_D = \left(\frac{k}{2c}\right)^{1/4}, \qquad T_D(q) = \frac{kh_q}{2} + \sqrt{2kc}.$$

Half of the target-independent cost is expected wage and half is record acquisition cost. With these two records, high effort is optimal under favorable evaluation exactly when

$$v[A_H(r) - A_L(r)] \geq \frac{k(2\alpha_H - 1)}{2} + \sqrt{2kc},$$

with low effort also optimal at equality. This compares the additional decision value with the optimized wage and acquisition costs.

The direct-record cost $T_D(q)$ is affine in q. Since $A(q; r)$ is convex, an interior target cannot improve on both pure choices under this record restriction. The favorable value using only the initial assessment and state records is therefore

$$V_D^F(r) = \max\{vA_L(r),\ vA_H(r) - T_D(\alpha_H)\}.$$

When high effort gives greater profit, the principal uses (x_D, y_D) and pays $k/(x_D y_D)$ on recorded agreement. When low effort gives greater profit, she pays zero and acquires no records.

Under the state-record price restriction in Proposition 2, $y_D = 1$ and $x_D = \min\{1, (k/(2c_0))^{1/3}\}$. The high-effort cost is

$$T_D(\alpha_H) = \frac{k(2\alpha_H - 1)}{2} + \frac{c_\theta}{2} + \begin{cases} k/2 + c_0/2, & c_0 \leq k/2, \\ \frac{3}{2} c_0^{1/3} (k/2)^{2/3}, & c_0 > k/2. \end{cases}$$

A.4 Initial and state record precision given decision-record precision

Fix z and a target q. Write $u = u_q$, $t = t_q$, $X = x + uz$, $Y = y + tz$, and $d_0 = 1 - (u^2 + t^2)z^2 > 0$. The ratio in the expected wage formula is

$$\frac{1 + uxz + tyz}{xy + txz + uyz} = \frac{d_0 + uzX + tzY}{XY - utz^2}.$$

Where the expected wage is finite, $XY > utz^2$. Expanding the denominator as a geometric series expresses this ratio as a positive sum of functions $X^{-a}Y^{-b}$ with $a, b \geq 0$. These functions are convex; their pointwise series limit is convex. Adding the positive quadratic acquisition cost makes the conditional objective strongly convex. At its unique minimizer, the first-order conditions are

$$c_0 x = \frac{k\{(y + t_q z)(1 + t_q yz) - u_q^2 z^2 y\}}{2K_q^2},$$
$$c_\theta y = \frac{k\{(x + u_q z)(1 + u_q xz) - t_q^2 z^2 x\}}{2K_q^2},$$

whenever the respective coordinate is interior. At a precision of one, the corresponding marginal wage saving is at least its marginal acquisition cost. In particular, both direct records are exact if and only if

$$c_0 \leq \frac{k}{2} \frac{1 + (t_q - u_q)z}{1 + (u_q + t_q)z}, \qquad c_\theta \leq \frac{k}{2} \frac{1 + (u_q - t_q)z}{1 + (u_q + t_q)z}. \tag{12}$$

Minimizing the resulting conditional value over $z \in [0,1]$ determines the joint precision choice.

The principal can also vary all three precisions in a fixed proportion. Fix $n = (n_0, n_\theta, n_d) \geq 0$ with $\max_j n_j = 1$ and $K_q(n) > 0$, and write $(x, y, z) = \sqrt{s}\, n$, where $0 < s \leq 1$ determines their level. Let $K_q(n)$ and $N_q(n)$ be the agreement-event coefficients evaluated at n, and put $c_n = c_0 n_0^2 + c_\theta n_\theta^2 + c_d n_d^2$. Total cost is

$$\frac{k}{2sK_q(n)} + \frac{k[N_q(n) - 1]}{2K_q(n)} + \frac{sc_n}{2}.$$

Differentiating with respect to s gives the unique minimum at

$$s^*(n) = \min\left\{1, \sqrt{\frac{k}{c_n K_q(n)}}\right\}.$$

For a high-effort guarantee, take a direction satisfying $\min\{K_L(n), K_H(n)\} > 0$. The same calculation replaces the incentive denominator $K_q(n)$ by $\min\{K_L(n), K_H(n)\}$ and evaluates the payment numerator at high effort.

A.5 The price at which the final-decision record is dropped

Fix high effort, and write $u_H = u_{\alpha_H}$, $t_H = t_{\alpha_H}$, and $T_D = T_D(\alpha_H)$. The derivative of the minimum expected wage at the optimized direct records is

$$\left.\frac{\partial W_S}{\partial z}\right|_{(x_D, y_D, 0)} = -\frac{k\{t_H x_D(1 - y_D^2) + u_H y_D(1 - x_D^2)\}}{2x_D^2 y_D^2}. \tag{13}$$

Here $c_S^*(r)$ is the supremum of prices $c_d > 0$ for which $T_S(\alpha_H; r) < T_D$, with value zero for an empty set.

Lemma A.1. *Fix the high-effort target, its Bayesian manager rule, and positive record prices. Then:*

1. *$T_S(\alpha_H; r) < T_D$ exactly when $0 < c_d < c_S^*(r)$. At every price $c_d > c_S^*(r)$, all record choices minimizing supporting cost have $z = 0$.*
2. *The price threshold is infinite exactly when*

$$t_H x_D(1 - y_D^2) + u_H y_D(1 - x_D^2) > 0. \tag{14}$$

Otherwise it is finite, possibly zero.

In particular, if the high-effort manager chooses $d = M_0$, the price threshold is finite exactly when $x_D = 1$.

Proof of Lemma A.1. The optimized cost is an infimum of functions affine and nondecreasing in the price vector, so it is concave and nondecreasing. Suppose one price increases from c to c' and its corresponding optimal centered precision changes from ξ to ξ'. Adding the two optimality inequalities gives $(c' - c)[(\xi')^2 - \xi^2] \leq 0$.

Direct evaluation is available at every c_d, so $T_S(\alpha_H; r) \leq T_D$. Strict improvement requires $z > 0$. The improving-price set is downward closed and open at every positive finite endpoint by continuity of the value function. If an optimal record choice had $z > 0$ at a price above the threshold, the same record choice at a slightly lower price still above the threshold would strictly improve on T_D. This is a contradiction.

For finiteness, let $F(x,y,z)$ be the objective in (2) at $q=\alpha_H$ with $c_d z^2/2$ removed. The exact threshold is

$$c_S^*(r)=\max\left\{0,\ \sup_{\substack{x,y\in[0,1]\\ z\in(0,1]}}\frac{2[T_D-F(x,y,z)]}{z^2}\right\}.$$

Write $\gamma_S=\partial_z F(x_D,y_D,0)$. If $\gamma_S<0$, evaluating this quotient at (x_D,y_D,z) and sending z to zero makes it unbounded. If $\gamma_S\geq 0$, strong convexity of the direct objective and a Taylor bound around its unique minimizer give positive constants a,b,c such that

$$F(x,y,z)\geq T_D+a\|(x,y)-(x_D,y_D)\|^2+\gamma_S z-b\|(x,y)-(x_D,y_D)\|z-cz^2.$$

The one-sided first-order terms at active upper bounds are nonnegative on the feasible cube. Completing the square bounds the right side below by $T_D+\gamma_S z-c'z^2$. Any improving sequence with $z\to 0$ approaches the unique direct optimum. Away from $z=0$ the threshold quotient is bounded, since $F\geq 0$. This proves finiteness. Substituting (13) gives (14); when $(u_H,t_H)=(m,0)$, its left side is positive exactly when $x_D<1$. □

A.6 The optimal contract without managerial review

When $d=M_0$, put $h_H=2\alpha_H-1$. A wage schedule minimizing expected payment while supporting high effort pays $2k/[y(x+mz)]$ when all three records agree. Its total cost, minimized over records, is

$$T_0=\min_{(x,y,z)\in[0,1]^3}\left\{\frac{k}{2}\left[h_H+\frac{1+mxz}{y(x+mz)}\right]+C(x,y,z)\right\}.$$

When $d=M_0$ under every effort conjecture, all coefficients except h_q in (2) are independent of q. Thus $T_S(q)=T_0+k(q-\alpha_H)$, and profit is affine in q on $(\alpha_L,\alpha_H]$, with slope $vm-k$. If this slope is positive, H yields the highest profit among these targets. If it is nonpositive, every target $q>\alpha_L$ yields less than $v\beta_L$: its decision gain is at most $vm(q-\alpha_L)$, and its expected wage is at least kq. Comparing with the zero-wage contract proves (6) for favorable evaluation. Multiplying an optimal supporting bonus by $1+\varepsilon$ makes effort strictly optimal under the fixed rule $d=M_0$, proving the robust equality in value.

In particular, if

$$c_0\leq\frac{k(1-m)}{2(1+m)},\qquad c_\theta\leq\frac{k}{2},\tag{15}$$

then exact direct records are optimal: $x=y=1,z=0$. The wage is k on $\widehat{a}_0=\widehat{\theta}$, and $T_0=k\alpha_H+(c_0+c_\theta)/2$.

At $u=m,t=0$, conditions (12) hold for every z whenever (15) holds. Conditional on $x=y=1$, the wage is kq and independent of z, so the positive acquisition price makes $z=0$ optimal. This gives the explicit contract above.

B Better managerial information

B.1 Record choice and the profit effect of better information

For $r>\beta(q)$, the threshold in (10) decreases strictly in r and increases strictly in q. Let $P_+(s)$ and $P_-(s)$ be the upper-tail probabilities of the normal components centered at $+1$ and -1, respectively:

$$P_+(s)=\Phi((1-s)/\tau),\qquad P_-(s)=\Phi((-1-s)/\tau).$$

The managerial coefficients are

$$u_q = m[1 - P_+(s_q) - P_-(s_q)], \qquad t_q = (2r-1)[P_+(s_q) - P_-(s_q)]. \tag{16}$$

Consequently u_q decreases strictly and t_q increases strictly with r whenever $r > \beta(q)$. They are constant in the remaining region. The envelope theorem gives the derivatives of equilibrium accuracy:

$$\frac{\partial A(q;r)}{\partial q} = u_q(r), \qquad \frac{\partial A(q;r)}{\partial r} = P_+(s_q) - P_-(s_q) > 0 \quad (r > \beta(q)).$$

Terms involving the derivative of the threshold cancel because the manager is indifferent at the switching observation. It follows that $u_q(r) \geq u_L(\bar{r}) > 0$ for all $q \in [\alpha_L, \alpha_H]$ and $r \leq \bar{r}$, and that $A_H(r) - A_L(r) = \int_{\alpha_L}^{\alpha_H} u_q(r)\, dq$ is nonincreasing in r.

The Blackwell comparison can also be verified directly. For $r_1 < r_2$, multiply an observation from the r_2 experiment by an independent random sign whose mean is $(2r_1 - 1)/(2r_2 - 1)$. Symmetry of the Gaussian noise gives the r_1 experiment. The distribution of M_0 conditional on a_0 is unchanged.

For fixed record precisions and a target q, suppress the target subscripts on u and t. With $x = 1$, the minimum expected wage is

$$W_S = \frac{k}{2}\left[h_q + \frac{1 + y\{tz/(1+uz)\}}{y + tz/(1+uz)}\right]. \tag{17}$$

As r increases, t rises and u falls, so $tz/(1+uz)$ rises. The wage is decreasing in this quantity when $y < 1$ and constant when $y = 1$. With $y = 1$, the corresponding expression is

$$W_S = \frac{k}{2}\left[h_q + \frac{1 + x\{uz/(1+tz)\}}{x + uz/(1+tz)}\right]. \tag{18}$$

Now $uz/(1+tz)$ falls with r. The wage consequently rises when $x < 1$, $z > 0$, and the manager's rule changes with r.

Lemma B.1. *For each target $q \in (\alpha_L, \alpha_H]$, every record choice that minimizes supporting cost has $x = 1$ if $c_0 \leq k(1-m)/[2(1+m)]$. For any $\bar{r} < 1$, every record choice that minimizes supporting cost has $y = 1$ at all $r \leq \bar{r}$ if $c_\theta < k(1-\bar{r})/(2\bar{r})$.*

Proof. Fix z and a managerial rule, and suppress target subscripts. At $y = 1$, the marginal wage saving from state-record precision is

$$-\partial_y W_S(x, 1, z) = \frac{k}{2}\frac{(x+uz)(1+uxz) - t^2 z^2 x}{[(1+tz)x + uz]^2}.$$

For $a \geq 0$, $0 \leq b < 1$, and $a + b < 1$, define

$$f(x; a, b) = \frac{(x+a)(1+ax) - b^2 x}{[(1+b)x + a]^2}.$$

On its finite domain,

$$\partial_x f = \frac{[a^2 - (1+b)^2][(1-b)x + a]}{[(1+b)x + a]^3} \leq 0.$$

Thus, with $a = uz$ and $b = tz$,

$$f(x; uz, tz) \geq f(1; uz, tz) = \frac{1 + uz - tz}{1 + uz + tz} \geq \frac{1-t}{1+t} \geq \frac{1-\bar{r}}{\bar{r}}.$$

The last inequality uses $t \leq 2r - 1 \leq 2\bar{r} - 1$. If the state price satisfies the stated condition, the derivative of $W_S + C$ at $y = 1$ is strictly negative for every x, z that can generate a positive incentive. The objective is strictly convex in y conditional on x, z. Its minimum therefore has $y = 1$.

Interchanging x, y and u, t gives

$$-\partial_x W_S(1, y, z) \geq \frac{k}{2}\frac{1-u}{1+u} \geq \frac{k}{2}\frac{1-m}{1+m}.$$

The inequality $u \leq m$ follows from (16). Strict convexity in x and the weak price inequality imply $x = 1$. If fixing the other coordinates makes the wage infinite for every positive precision, that pair of coordinates cannot be part of an optimal record choice and can be omitted from the argument. □

Under the first condition of Lemma B.1, (17) shows that the wage at every formerly optimal record choice weakly decreases as r increases. Acquisition costs are unchanged. Hence $T_S(q; r)$ is nonincreasing at each target. Taking the supremum over $q \in (\alpha_L, \alpha_H]$ and comparing with the zero-wage alternative proves the global nondecrease of V^F asserted after (4).

B.2 Changes in initial and state record precision

Fix $q \in (\alpha_L, \alpha_H]$, prices $c_0, c_\theta, c_d > 0$, and all other primitives except r. Consider $r_1 < r_2$ with $r_2 > \beta(q)$. For $i = 1, 2$, write $u_i = u_q(r_i)$ and $t_i = t_q(r_i)$, and let (x_i, y_i, z_i) be any minimizer of $T_S(q; r_i)$. Then $u_2 < u_1$ and $t_2 > t_1$. If every minimizer at both signal precisions has $y = 1$, then $x_2 \geq x_1$. If every minimizer at both signal precisions has $x = 1$, then $y_2 \leq y_1$. When $r_1 < r_2 \leq \beta(q)$, the objectives coincide and the optimal sets are equal.

Proof. First suppose all minima have $y = 1$. For a fixed pair $u = u_q(r) > 0$ and $t = t_q(r) \geq 0$, use the change of variable

$$\eta = \frac{uz}{1+tz}, \qquad z = \frac{\eta}{u - t\eta}, \qquad 0 \leq \eta \leq \frac{u}{1+t}.$$

The denominator is positive on this domain. Up to the constant $kh_q/2 + c_\theta/2$, the cost is

$$F(x, \eta) + Q(\eta; u, t), \qquad F(x, \eta) = \frac{k}{2}\frac{1 + x\eta}{x + \eta} + \frac{c_0 x^2}{2}, \qquad Q(\eta; u, t) = \frac{c_d}{2}\left(\frac{\eta}{u - t\eta}\right)^2.$$

At $x = \eta = 0$ assign infinite cost. Since $u + t < 1$, every feasible η is less than one. Conditional on η, the minimizer $x(\eta) \in [0, 1]$ is unique, because

$$F_{xx} = c_0 + \frac{k(1 - \eta^2)}{(x + \eta)^3} > 0, \qquad F_{x\eta} = \frac{k(1 + x\eta)}{(x + \eta)^3} > 0.$$

The conditional minimizer is therefore nonincreasing in η, including where it equals one.

Let $F_*(\eta) = \min_{x \in [0,1]} F(x, \eta)$ and $Q_i(\eta) = Q(\eta; u_i, t_i)$. The remaining problem minimizes $F_*(\eta) + Q_i(\eta)$ on $[0, u_i/(1 + t_i)]$. Its acquisition marginal cost is

$$Q_i'(\eta) = \frac{c_d u_i \eta}{(u_i - t_i\eta)^3}.$$

For $\eta > 0$, decreasing u_i and increasing t_i strictly raise this marginal cost. Thus $Q_2 - Q_1$ is strictly increasing on the common domain, and the upper feasible bound falls. Suppose two minimizers

satisfied $\eta_2 > \eta_1$. Both choices would be feasible at both signal precisions. Their optimality inequalities imply

$$F_*(\eta_1) + Q_1(\eta_1) \le F_*(\eta_2) + Q_1(\eta_2), \qquad F_*(\eta_2) + Q_2(\eta_2) \le F_*(\eta_1) + Q_2(\eta_1).$$

Adding gives $(Q_2 - Q_1)(\eta_2) \le (Q_2 - Q_1)(\eta_1)$, a contradiction. Hence $\eta_2 \le \eta_1$, and the unique conditional choice gives $x_2 = x(\eta_2) \ge x(\eta_1) = x_1$.

Now suppose all minima have $x = 1$. If $t > 0$, set

$$\eta = \frac{tz}{1+uz}, \qquad z = \frac{\eta}{t - u\eta}, \qquad 0 \le \eta \le \frac{t}{1+u}.$$

The same conditional problem applies to y, replacing c_0 by c_θ. Its unique minimizer $y(\eta)$ is nonincreasing in η. The final-record acquisition cost now has marginal cost

$$\frac{c_d t \eta}{(t - u\eta)^3}.$$

Increasing t and decreasing u strictly lower this marginal cost at positive η, while expanding the feasible interval. The two optimality inequalities consequently rule out $\eta_2 < \eta_1$. It follows that $y_2 \le y_1$. If $t_1 = 0$, the wage at r_1 is independent of z, so $z_1 = \eta_1 = 0$. Then $\eta_2 \ge 0$ directly implies $y_2 \le y(0) = y_1$. Finally, where the manager follows M_0 both coefficients are constant, so the entire objective and its optimal set are unchanged. □

If exact state recording is optimal at both signal precisions, then $T_S(q; r_2) \ge T_S(q; r_1)$. Equation (18) orders wages at every record choice with $y = 1$, and acquisition costs are unchanged. Taking the minima over that common set preserves the ordering. The corresponding argument with (17) gives the reverse cost ordering when the initial record is exact at both optima.

B.3 Changes in final-decision record precision

Fix $q \in (\alpha_L, \alpha_H]$, positive acquisition prices, and all other primitives except r. Compare $r_1 < r_2$ with $r_2 > \beta(q)$, and let (x_i, y_i, z_i) be any record choice minimizing supporting cost at r_i. If every minimizer at both signal precisions has $y = 1$ and $c_0 \le 2k$, then $z_2 \le z_1$. If every minimizer at both signal precisions has $x = 1$ and $c_\theta \le 2k$, then $z_2 \ge z_1$.

Proof. The two cases share a conditional cost problem. With an exact state record, set $(\ell, a, b, c) = (x, u_q, t_q, c_0)$; with an exact initial record, set $(\ell, a, b, c) = (y, t_q, u_q, c_\theta)$. Define

$$\eta = \frac{az}{1+bz}, \qquad g(\eta) = \min_{0<\ell\le 1} \left\{ \frac{k}{2} \frac{1+\ell\eta}{\ell+\eta} + \frac{c\ell^2}{2} \right\}.$$

Here $a, b \ge 0$, $a + b < 1$, and $\eta < 1$. The conditional minimizer $\ell(\eta)$ is unique and positive by the strict convexity established in Appendix B.2. At an interior solution,

$$c\ell = \frac{k(1-\eta^2)}{2(\ell+\eta)^2}, \qquad \ell'(\eta) = -\frac{2\ell(1+\ell\eta)}{(1-\eta^2)(3\ell+\eta)}.$$

The marginal saving from η is

$$M(\eta) = -g'(\eta) = \frac{k[1-\ell(\eta)^2]}{2[\ell(\eta)+\eta]^2}.$$

It is positive when $\ell < 1$ and zero when the direct record is exact.

The price bound $c \le 2k$ implies $M + \eta M' > 0$ wherever $\ell < 1$. For $\eta = 0$ this follows from $M > 0$. For $0 < \eta < 1$, the first-order condition gives

$$\frac{1-\eta^2}{\ell(\ell+\eta)^2} = \frac{2c}{k} \le 4.$$

The left side is strictly decreasing in ℓ and equals $8/(1+\eta) > 4$ at $\ell = (1-\eta)/2$. Hence $\ell > (1-\eta)/2$. Differentiation gives

$$M + \eta M' = M \frac{P(\ell, \eta)}{(1-\eta^2)(3\ell+\eta)(1-\ell^2)},$$

where

$$P(\ell, \eta) = (1+\eta^2)(3\ell+\eta)(1-\ell^2) - 2\eta(1+\ell\eta)(1-3\ell^2).$$

For fixed η, its derivative is

$$P_\ell = -9(1-\eta^2)\ell^2 + 2\eta(5-\eta^2)\ell + 3 + \eta^2.$$

This quadratic has one positive and one negative root, so P first increases and then decreases on the positive half-line. Its minimum on $[(1-\eta)/2, 1]$ occurs at an endpoint, and

$$P((1-\eta)/2, \eta) = \frac{(1-\eta)^2(1+\eta)(9+8\eta-5\eta^2)}{8} > 0,$$
$$P(1, \eta) = 4\eta(1+\eta) > 0.$$

This proves the claimed inequality and also gives $2M + \eta M' > 0$. On an interval with $\ell = 1$, $M = 0$ and both weak inequalities hold. The marginal saving is continuous across this boundary, so the derivative comparisons below extend across it by integration.

After optimizing the direct record, the cost that varies with z is

$$J(z; a, b) = g\left(\frac{az}{1+bz}\right) + \frac{c_d z^2}{2}.$$

At differentiability points, its derivatives satisfy

$$J_z = c_d z - M(\eta)\frac{a}{(1+bz)^2},$$
$$J_{za} = -\frac{M + \eta M'}{(1+bz)^2} \le 0,$$
$$J_{zb} = \frac{az}{(1+bz)^3}(2M + \eta M') \ge 0.$$

The last two inequalities are strict when $a > 0$, $z > 0$, and $M > 0$. With an exact state record, $a = u_q$ falls and $b = t_q$ rises as r increases. Thus the new cost minus the old cost is nondecreasing in z.

Any global optimum with $z > 0$ has $\ell < 1$: at $\ell = 1$ the wage is independent of z, and setting $z = 0$ would save its positive acquisition cost. Suppose $z_2 > z_1$. At z_2, the new conditional optimum therefore has $M > 0$. The old value of η at this same z_2 is larger, so its conditional direct precision is also below one. The cost difference is strictly increasing on an interval just below z_2. Writing $J_i(z) = J(z; a_i, b_i)$, global optimality gives

$$J_1(z_1) \le J_1(z_2), \qquad J_2(z_2) \le J_2(z_1),$$

and hence $(J_2 - J_1)(z_2) \leq (J_2 - J_1)(z_1)$, a contradiction. This proves $z_2 \leq z_1$, including boundary and nonunique optima.

With an exact initial record, $a = t_q$ rises and $b = u_q$ falls. The cost difference is nonincreasing in z. If $z_1 > z_2$, positive use at the old optimum z_1 supplies the same strictness, contradicting the two optimality inequalities. Thus $z_2 \geq z_1$. If $t_q(r_1) = 0$, the old wage is independent of z and its optimum has $z_1 = 0$, which gives the result directly. □

The opposite adjustment can occur when the initial record is expensive. Fix the same target and information comparison, set $u_i = u_q(r_i)$, and suppose

$$c_\theta < \frac{k(1 - r_2)}{2r_2}, \qquad c_d > \frac{k}{2u_2}.$$

The first condition makes state recording exact at every optimum. Hold these prices fixed and let c_0 increase. Then every optimal record choice satisfies

$$x_i \longrightarrow 0, \qquad z_i \longrightarrow \left(\frac{k}{2c_d u_i}\right)^{1/3} < 1.$$

Since $u_2 < u_1$, for all sufficiently large finite c_0, every optimal final-decision record precision at r_2 strictly exceeds every optimal final-decision record precision at r_1.

To establish the limit, a feasible record choice with $x = 0$, $y = 1$, and $z > 0$ bounds cost independently of c_0. Every optimizing sequence therefore has $x \to 0$, while z stays bounded away from zero because the wage diverges as the incentive denominator vanishes. At $x = 0$, the only terms depending on z are

$$\frac{k}{2u_i z} + \frac{c_d z^2}{2}.$$

This function is strictly convex and has the unique interior minimizer displayed above. Comparing any limit point with this feasible record choice shows that all optimal final-decision record precisions converge to that minimizer. The strictly ordered limits imply the ordering at sufficiently large finite initial-record prices. The finite-price problem continues to optimize all three record precisions.

B.4 The profit effect when the manager begins to revise

Proof of Proposition 2. Lemma B.1 gives $y = 1$ at all record choices minimizing supporting cost in the stated range. First take a price $0 < c_d < c_S^*(\beta_H)$. At $r = \beta_H$, the minimum supporting cost is strictly below the direct-record cost, so every optimal record choice has $z > 0$. If also $x = 1$, the wage in (18) is independent of z. Reducing z to zero then lowers acquisition cost without changing the wage. Hence an optimal record choice with $z > 0$ has $x < 1$. At $x = 0$ and $z > 0$, the wage derivative with respect to x is strictly negative because $0 < uz/(1 + tz) < 1$, whereas the acquisition derivative is zero. Thus $0 < x < 1$ at each of these record choices.

For fixed record precisions with $y = 1$, differentiation with respect to the target gives

$$\partial_q \left(\frac{u_q z}{1 + t_q z}\right) = \frac{z u_q' [1 + t_q z + h_q u_q z]}{(1 + t_q z)^2} \geq 0,$$

where $t_q' = -h_q u_q'$. Since for $\xi \geq 0$, $\frac{\partial}{\partial \xi}[(1 + x\xi)/(x + \xi)] = (x^2 - 1)/(x + \xi)^2 \leq 0$, $\partial_q W_S \leq k$. The same bound holds on an interval where the manager follows M_0 and across its continuous boundary. Taking an optimal record choice at a lower target and applying it at a higher target therefore gives

$$T_S(q_2; r) - T_S(q_1; r) \leq k(q_2 - q_1) \quad (\alpha_L < q_1 < q_2 \leq \alpha_H).$$

Equilibrium accuracy increases by at least $u_L(\bar{r})(q_2 - q_1)$. The first bound on v makes $vA(q;r) - T_S(q;r)$ strictly increasing on $(\alpha_L, \alpha_H]$. Moreover,

$$v[A_H(r) - A_L(r)] \geq v[A_H(\bar{r}) - A_L(\bar{r})] > T_D(\alpha_H) \geq T_S(\alpha_H; r).$$

This proves strict optimality of high effort in the complete favorable problem for every $c_d > 0$. With $y = 1$ and $z = 0$, minimizing $k/(2x) + c_0 x^2/2$ on $(0, 1]$ gives $x_D = \min\{1, (k/(2c_0))^{1/3}\}$, which also establishes the displayed direct-cost formula.

For the local information comparison, put $h = 2\alpha_H - 1$ and let $r \downarrow \beta_H$ from above. Then $s_H \to \infty$. Gaussian tail ratios satisfy $P_-(s_H)/P_+(s_H) \to 0$. With $\varepsilon = m - u_H = m[P_+(s_H) + P_-(s_H)]$,

$$\frac{t_H}{\varepsilon} = \frac{2r-1}{m} \frac{P_+(s_H) - P_-(s_H)}{P_+(s_H) + P_-(s_H)} \longrightarrow h.$$

Thus $t_H = h\varepsilon + o(\varepsilon)$, and $A_H - \beta_H = (t_H - h\varepsilon)/2 = o(\varepsilon)$. The derivative of equilibrium accuracy above makes this change strictly positive.

Let $\mathcal{M}_c \subset [0,1]^3$ be the set of centered precision triples minimizing supporting cost at $r = \beta_H$, and set $K_c = y(x + mz)$. Expanding the wage at a fixed precision vector gives

$$W_S(\alpha_H; x, y, z; r) - W_S(\alpha_H; x, y, z; \beta_H) = -\varepsilon \Gamma(x, y, z) + o(\varepsilon),$$

where

$$\Gamma(x, y, z) = \frac{kz}{2K_c^2} \left[h(x + mz)(1 - y^2) - (y + hmz)(1 - x^2) \right].$$

The expansion is uniform on a compact neighborhood of the optimal set with K_c bounded away from zero. Such a neighborhood contains every optimizer for sufficiently close r: direct evaluation gives a uniform total-cost bound, while $W_S \geq k/(2K_q)$ excludes sequences with $K_q \to 0$.

Uniform expansion and compactness give the optimized-value formula

$$[vA_H(r) - T_S(\alpha_H; r)] - [v\beta_H - T_S(\alpha_H; \beta_H)] = \varepsilon \max_{(x,y,z) \in \mathcal{M}_c} \Gamma(x, y, z) + o(\varepsilon). \tag{19}$$

The upper bound follows by evaluating each new optimizer in the old problem and taking a convergent subsequence. Its limit belongs to $\mathcal{M}_c$. The lower bound follows by retaining an old optimizer that maximizes Γ. The contribution from equilibrium accuracy is $o(\varepsilon)$ because v is fixed.

At a price below $c_S^*(\beta_H)$, every $(x, y, z) \in \mathcal{M}_c$ has $y = 1$, $x < 1$, and $z > 0$, so

$$\Gamma(x, y, z) = -\frac{kz(1 - x^2)(1 + hmz)}{2(x + mz)^2} < 0.$$

Continuity and compactness imply $\max_{(x,y,z) \in \mathcal{M}_c} \Gamma(x, y, z) < 0$. Equation (19), together with high effort's strict optimality, proves the asserted decrease of V^F.

If $c_d \geq c_S^*(\beta_H)$, direct comparison is optimal at $r = \beta_H$. For a positive finite threshold, this includes equality by continuity of the optimized cost in c_d. Its wage and acquisition cost are independent of the managerial rule. Retaining the optimal assessment and state records after the information improvement gives

$$V^F(r) \geq vA_H(r) - T_D(\alpha_H) > v\beta_H - T_D(\alpha_H) = V^F(\beta_H) \quad (\beta_H < r \leq \bar{r}).$$

Finally, if $c_0 > k/2$, then $x_D < 1$. The optimal assessment and state records give

$$\partial_z (W_S + C)|_{(x_D, 1, 0),\, r = \beta_H} = -\frac{km(1 - x_D^2)}{2x_D^2} < 0.$$

This strict first-order wage saving dominates the quadratic acquisition cost of a sufficiently small z at every finite c_d. Thus the direct record choice never minimizes supporting cost, and $c_S^*(\beta_H) = \infty$. □

To verify the positive finite price interval in (5), let $a = c_0/k$. The direct optimum has $x = y = 1$. At the auxiliary price $c_d = 0$, taking $y = z = 1$ and $x = 1$ gives exactly the direct cost. The objective's derivative in x there is

$$c_0 - \frac{k}{2}\frac{1-m}{1+m} > 0.$$

Reducing x therefore saves total cost. The strict saving persists for sufficiently small positive c_d, proving $c_S^*(\beta_H) > 0$.

For finiteness, define

$$z_0 = \frac{1-2a}{m(1+2a)} \in (0,1).$$

At $r = \beta_H$ and $y = 1$, the objective conditional on z is strictly convex in x. Its derivative at $x = 1$ is nonpositive for $z \le z_0$, so its conditional minimum has $x = 1$. Such a record choice has the direct wage and the additional positive cost $c_d z^2/2$. For $z \ge z_0$, acquisition cost alone exceeds $T_D(\alpha_H)$ when $c_d > 2T_D(\alpha_H)/z_0^2$. Hence direct evaluation is optimal at every sufficiently large c_d, and $c_S^*(\beta_H) < \infty$.

B.5 The information threshold for reducing effort

Proof of Proposition 3. If the manager follows $\text{sign}(S)$, her decision is correct with probability $r - (2r-1)\Phi(-1/\tau)$, so

$$A_L(r) \ge r - (2r-1)\Phi(-1/\tau).$$

At any target with $\beta(q) \le r$, even observing the ungarbled pair (M_0, Z_r) gives a probability of a correct decision of at most r: the more accurate binary signal is Z_r. Hence

$$v[A(q;r) - A_L(r)] \le v(2r-1)\Phi(-1/\tau) < k\alpha_L < kq.$$

Every positive-effort equilibrium requires an expected wage of at least kq. Such targets consequently yield less profit than zero wages and zero acquisition, including at $\beta(q) = r$.

If an interior target satisfies $\beta(q) > r$, the manager follows M_0 throughout the interval from q to α_H. Analyst indifference at the interior target implies that all targets in this interval are equilibria of the same contract. Per unit of initial assessment accuracy, the expected wage difference is k and the equilibrium accuracy difference is m. The principal's equilibrium profit has slope $vm - k$. If the slope is positive, high effort gives strictly higher profit. If the slope is nonpositive, the additional decision value over low effort is at most $vm(q - \alpha_L)$, which is smaller than the required wage kq. Zero wages again give strictly higher profit. All interior targets have therefore been excluded from a favorable optimum.

For $r \le \beta_H$, high effort has equilibrium accuracy β_H and minimum total supporting cost T_0. The full favorable value is

$$V^F(r) = \max\{vA_L(r),\ v\beta_H - T_0\}.$$

For $r \ge \beta_H$, the first argument applies to every target $q > \alpha_L$, and zero wages are strictly optimal.

The function $A_L(r)$ equals β_L for $r \le \beta_L$ and is continuous and strictly increasing thereafter. By assumption, $v(\beta_H - \beta_L) > T_0$. At the other endpoint,

$$v[\beta_H - A_L(\beta_H)] \le v(2\beta_H - 1)\Phi(-1/\tau) < k\alpha_L < T_0,$$

where $T_0 \ge k\alpha_H$ follows from the wage bound. Continuity and strict monotonicity yield the unique solution $r^* \in (\beta_L, \beta_H)$ to (7). The value formula, effort choices, and equilibrium accuracy difference follow. At the threshold, $\beta_H - A_L(r^*) = T_0/v$, so switching from high to low effort lowers equilibrium accuracy by T_0/v. □

C Contracts under robust evaluation

C.1 The cost of guaranteeing high effort

Derivation of the guarantee cost. Every contract guaranteeing H satisfies $D_w(\alpha_L) > k$ and $D_w(\alpha_H) \geq k$. Relax the first requirement to a weak inequality. At a fixed precision vector, averaging payments across sign reversals preserves the high-effort wage bill and both incentive requirements. The resulting wage problem is a linear program on four record classes, indexed by $(\widehat{a}_0\widehat{\theta}, \widehat{a}_0\widehat{d})$. It has an optimum with at most two positive payments.

Any two distinct classes are interchanged by reversing one recorded coordinate. If their payments are $a \geq b \geq 0$, reduce that coordinate's centered precision from ξ to $\xi(a-b)/(a+b)$ and pay $a+b$ on the first class. Conditional record probabilities are affine in ξ. The new wage therefore has the same expected payment conditional on each realization (a_0, θ, d), preserves incentives at every conjecture, and weakly reduces acquisition cost.

A wage paying on one remaining class can be reduced to a bonus for agreement among all three records. For the class $(+1, -1)$, the incentive coefficient is $xy - t_q xz - u_q yz$, and

$$\frac{xy - t_H xz - u_H yz}{1 - u_H xz - t_H yz} \leq xy.$$

Deleting the final-decision record and using the direct bonus $k/(xy)$ weakly reduces the high-effort wage bill and satisfies both constraints. For the class $(-1, +1)$, the coefficient $t_q xz - xy - u_q yz$ decreases with q, so the high endpoint binds. At $q = \alpha_H$, its ratio K_{sj}/L_{sj} decreases in y: the derivative has numerator

$$-(x + u_H z)(1 + u_H xz) + t_H^2 xz^2 < 0.$$

Deleting the state record and rescaling the bonus improves the binding incentive per unit of expected payment; monotonicity preserves the incentive constraint at the low-effort expectation.

For the class $(-1, -1)$, the coefficient $u_q yz - xy - t_q xz$ increases with q, so the low endpoint binds. Put $a_H = t_H + h_H u_H$ and $b_H = u_H + h_H t_H$. The high-effort wage ratio is

$$\frac{1 + a_H yz - x(h_H y + b_H z)}{u_L yz - x(y + t_L z)}.$$

Its derivative in x has numerator

$$(y + t_L z)(1 + a_H yz) - (h_H y + b_H z)u_L yz \geq y(1 - u_L u_H z^2) + t_L z > 0.$$

Deleting the initial record lowers the wage bill. After the deletion in each of the last two cases, the remaining event is equivalent in distribution to agreement among all three records. Each precision reduction lowers acquisition cost, so these comparisons establish the lower bound over all wage schedules and record choices.

For a bonus paid when all three records agree, K_q is concave as a function of u_q. Its minimum is attained at an endpoint, giving (8). The relaxed minimum is therefore exactly (9). Multiplying the minimizing bonus by $1 + \varepsilon$ makes $D_w(q) > k$ for all q and guarantees H. The associated costs converge to the relaxed value. The numerator of the wage ratio is bounded below by a positive constant on the cube; cost consequently diverges at a zero denominator. Compactness gives a minimizing precision vector.

Pointwise comparison with the supporting objective gives $T_S \leq T_G$, and direct evaluation gives $T_G \leq T_D$. A record choice minimizing supporting cost with $K_L \geq K_H$ also minimizes the expression in (9). Conversely, when the two values coincide, any minimizer of that expression must also minimize supporting cost, and its two denominators must agree. Thus $K_L \geq K_H$ there. □

C.2 Optimal effort under robust evaluation

Let $T_I(q;r)$ be the infimum total cost of contracts whose only continuation equilibrium has initial assessment accuracy $q \in (\alpha_L, \alpha_H)$, with $\beta(q) < r$.

Lemma C.1. *For $q \in (\alpha_L, \alpha_H)$ with $\beta(q) < r$, the infimum total cost of unique implementation is*

$$T_I(q;r) = \min_{\substack{(x,y,z)\in[0,1]^3 \\ K_L \geq K_q}} \left\{ \frac{kN_q}{2K_q} + C(x,y,z) \right\}. \tag{20}$$

A zero denominator is assigned infinite cost. The optimal robust value is

$$\begin{aligned} V^R(r) = \max \Big\{ & vA_L(r),\ vA_H(r) - T_G(r), \\ & \sup_{\substack{q\in(\alpha_L,\alpha_H) \\ \beta(q)<r}} [vA(q;r) - T_I(q;r)] \Big\}. \end{aligned} \tag{21}$$

If no interior target satisfies the stated restrictions, the supremum term is omitted.

The inequality $K_L \geq K_q$ is the weak form of the incentive requirement at the low-effort expectation. When it binds, the proof uses a contract perturbation to exclude the competing low-effort equilibrium while approaching the stated cost.

Proof. For the interior target q, relax the wage problem to $D_w(q) = k$ and $D_w(\alpha_L) \geq k$. Its fixed-precision linear program has an optimum with at most two positive payments. The reduction of two payments to one by adjusting record precision in the preceding proof preserves both constraints. For a wage paying only when all three records agree, the constraint is $K_L \geq K_q$ in (20). The increasing class $(-1,-1)$ cannot satisfy the incentive inequality at the low-effort expectation and equality at the target in the region $\beta(q) < r$. For the decreasing class $(-1,+1)$, deleting the state record improves incentive efficiency at the target; the resulting assessment–decision agreement has its effort-induced probability difference decreasing in the manager's expected initial assessment accuracy and hence satisfies the incentive inequality at the low-effort expectation. For $(+1,-1)$, direct comparison lowers target wages and has a constant effort-induced probability difference. This establishes the lower bound in (20) over all wages and precisions.

Consider first a minimizing record choice in (20) with $K_L > K_q$. Strict concavity of t in u where $\beta(q) < r$ implies $K_s > K_q$ for $\alpha_L \leq s < q$ and $K_s < K_q$ for $q < s \leq \alpha_H$. The bonus $2k/K_q$ uniquely implements q. At a weak minimum with $K_L = K_q$ and $z > 0$, the identity

$$K_L - K_q = z[(t_L - t_q)x - (u_q - u_L)y]$$

shows that a small reduction in y makes the inequality strict. Both x and y are positive in this equality case: otherwise the strict variation of t or u would violate either $K_q > 0$ or the equality. Rescale the bonus to retain target indifference. Costs converge to the weak minimum.

If the minimizing record choice has $z = 0$, then $xy > 0$. For $\varepsilon > 0$, acquire $z = \varepsilon$ and use

$$w_\varepsilon(R) = b_D \mathbf{1}\{\widehat{a}_0 = \widehat{\theta}\} + b_A \mathbf{1}\{\widehat{a}_0 = \widehat{d}\}, \qquad b_D = \frac{k - b_A t_q x \varepsilon}{xy},$$

where $b_A > 0$ tends to zero with ε. These payments are nonnegative for sufficiently small ε. At conjecture s,

$$D_{w_\varepsilon}(s) = k + b_A x \varepsilon (t_s - t_q).$$

It exceeds k below q and is below k above q, so the wage uniquely implements the target and approaches the direct cost. This establishes the infimum characterization even when the optimal assessment and state records solve the weak problem.

A contract with a low-effort equilibrium has robust profit at most $vA_L(r)$. If it excludes low effort, its incentive there is strict. Each interior equilibrium with $\beta(q) < r$ is then bounded by the corresponding relaxed cost (20). If its only equilibrium has high effort, its profit is bounded by $vA_H - T_G$. The remaining case has all its equilibrium targets in a range where $d = M_0$, including an interior target. Analyst indifference throughout that range implies a high-effort equilibrium. Continuity and the absence of equilibria below the range where the manager follows M_0 imply $D_w(s) > k$ there. A small multiplicative increase in all payments then guarantees H, and its high-effort wage bill converges to that of the original contract. The original robust payoff is at most its high-effort payoff and hence at most $vA_H - T_G$. These cases prove the upper bound in (21). Zero wages, the high-effort guarantees, and the unique interior implementations prove the reverse bound. □

C.3 Decision-record prices and the cost of guaranteeing high effort

Write

$$F_G(x, y, z) = \frac{kN_H}{2\min\{K_L, K_H\}} + \frac{c_0 x^2 + c_\theta y^2}{2}.$$

This objective excludes the acquisition cost of the final-decision record. The pair (x_D, y_D) minimizes $F_G(x, y, 0)$ uniquely, with value $T_D(\alpha_H)$.

Let $\gamma_G = \partial_z^+ F_G(x_D, y_D, 0)$ be the right derivative with respect to final-decision record precision at the direct optimum. With $h_H = 2\alpha_H - 1$, it equals

$$\begin{aligned}\gamma_G = \frac{k}{2x_D y_D}\Big[&(u_H + h_H t_H)x_D + (t_H + h_H u_H)y_D \\ &- \left(h_H + \frac{1}{x_D y_D}\right)\min_{i\in\{L,H\}}(t_i x_D + u_i y_D)\Big].\end{aligned} \tag{22}$$

Lemma C.2. *For the high-effort guarantee problem, the price threshold is*

$$c_G^*(r) = \max\left\{0, \sup_{\substack{(x,y,z)\in[0,1]^3 \\ z>0}} \frac{2[T_D(\alpha_H) - F_G(x, y, z)]}{z^2}\right\}. \tag{23}$$

It is infinite exactly when $\gamma_G < 0$, with γ_G given by (22). If $\gamma_G > 0$ and $0 < c_G^*(r) < \infty$, precision z in strictly improving optimal record choices stays bounded away from zero as c_d increases to $c_G^*(r)$. Moreover, $c_G^*(r) \le c_S^*(r)$.

Proof. A record choice with $z > 0$ strictly improves on direct evaluation exactly when its threshold ratio in (23) exceeds c_d. This proves the formula and the strict improving-price characterization. If an optimal record choice had $z > 0$ at a price strictly above $c_G^*(r)$, decreasing the price slightly while remaining above the threshold would produce a strict improvement, a contradiction. The ordering follows from $T_S \le T_G$.

At $z = 0$, the two endpoint expressions for expected wages agree. The right derivative of their maximum is the larger of their derivatives. Differentiating gives (22). If it is negative, evaluating (23) at (x_D, y_D, z) and letting z decrease to zero makes the ratio arbitrarily large.

If $\gamma_G \geq 0$, take an endpoint expression for expected wages whose right derivative is γ_G. In a neighborhood of the direct optimum, smoothness and strong convexity of the direct objective give constants $a, b, c > 0$ with

$$F_G(x, y, z) \geq T_D + a\|(x, y) - (x_D, y_D)\|^2 + \gamma_G z - b\|(x, y) - (x_D, y_D)\|z - cz^2.$$

At an upper precision bound of one, the first-order optimality inequality contributes a nonnegative term, so the same lower bound holds on the feasible cube. Completing the square gives $F_G \geq T_D + \gamma_G z - c'z^2$. Every sequence with $z \to 0$ and $F_G < T_D$ approaches the unique direct minimizer. The threshold ratio is consequently bounded near $z = 0$. Away from zero it is bounded above by $2T_D/z^2$, which proves finiteness. If $\gamma_G > 0$, the local bound excludes strict improvement for all sufficiently small positive z, proving that optimal positive precision stays bounded away from zero. □

C.4 Record choice under a high-effort guarantee

Lemma C.3. *In the benchmark, managerial state evidence has precision at most β_L, and the received assessment is executed at every effort expectation. After the improvement, precision is $r \in (\beta_L, \beta_H)$. All other primitives are unchanged. Let (x_D, y_D) minimize the high-effort cost using only initial-assessment and state records, and write that cost as $T_D(\alpha_H) = W_D + C_D$. Suppose $x_D, y_D < 1$ and*

$$m(x_D + h_H y_D) > \left(h_H + \frac{1}{x_D y_D}\right)(t_L x_D + u_L y_D), \tag{24}$$

where u_L, t_L describe the manager's rule at the low-effort expectation after the improvement. Suppose also

$$v(\beta_H - r) > T_D(\alpha_H). \tag{25}$$

The threshold $c_G^*(r) = \sup\{c_d > 0 : T_G(r) < T_D(\alpha_H)\}$, with value zero for an empty set, is finite. For every finite price $c_d > c_G^*(r)$:

1. *In the benchmark, every minimum-cost record choice supporting high effort uses an informative final-decision record. Write its total cost as $T_0 = W_0 + C_0$.*

2. *After the improvement, $(x_D, y_D, 0)$ minimizes the expression in (9), giving*

$$T_G(r) = T_D(\alpha_H) > T_0, \qquad C_D > C_0.$$

3. *High effort is optimal under favorable evaluation at both information precisions. The robust values are approached by contracts guaranteeing high effort. Equilibrium accuracy is β_H in these outcomes, and*

$$V_0 = V^F(r) = v\beta_H - T_0, \qquad V^R(r) = v\beta_H - T_D(\alpha_H),$$

where V_0 is the benchmark value under either evaluation.

When the benchmark without review chooses high effort, direct evaluation also gives the bound

$$V_0 - V^R(r) \leq T_D(\alpha_H) - T_0 - v[A_H(r) - \beta_H]. \tag{26}$$

Proof of Lemma C.3. In the stated information interval, $(u_H, t_H) = (m, 0)$, so the high-effort supporting cost equals T_0. The high-endpoint derivative in the guarantee objective at the direct optimum is $-km(1-x_D^2)/(2x_D^2 y_D) < 0$. Condition (24) says that the low-endpoint derivative is strictly positive. Hence $\gamma_G > 0$, and Lemma C.2 gives a finite guarantee price threshold. The negative supporting derivative makes $T_0 < T_D$ at every finite c_d, yielding the stated strict cost comparison.

A target with $\beta(q) \le r$ has $A(q;r) \le r$, so its profit is at most vr. Condition (25) gives a direct high-effort profit $v\beta_H - T_D > vr$. On the remaining range, the manager follows M_0 and minimum supporting cost has slope k in q. The same condition implies $vm > k$, since $T_D \ge k\alpha_H$ and $\beta_H - r < m\alpha_H$. Favorable profit is therefore increasing on this range, and high effort is the global optimum. For robust evaluation, a contract with a mixed equilibrium in the range where the manager follows M_0 also has an equilibrium at the lower boundary of that range or below: either low effort or a target satisfying $\beta(q) \le r$. Such a contract has value at most vr. Therefore every contract whose robust value exceeds vr guarantees H. This proves the complete value comparison and also shows that every sufficiently near-optimal robust contract guarantees H.

At the interior direct optimum, the first-order conditions give

$$C_D = c_0 x_D^2 = c_\theta y_D^2 = \frac{k}{2x_D y_D}, \qquad T_D - \frac{kh_H}{2} = 2C_D.$$

At any benchmark minimizer, feasible radial contraction implies $C_0 \le k/(2K_H)$. The expected wage expression implies $W_0 - kh_H/2 \ge k/(2K_H)$. Therefore

$$2C_0 \le T_0 - \frac{kh_H}{2} < T_D - \frac{kh_H}{2} = 2C_D.$$

Finally, direct high-effort guarantees are available under review at costs approaching T_D. Thus $V^R(r) \ge vA_H(r) - T_D$. Subtracting from $V_0 = v\beta_H - T_0$ proves (26). □

C.5 Expected wages and record acquisition costs

Lemma C.4. *Suppose the optimal direct precisions $X = x_D$ and $Y = y_D$ lie in $(0,1)$. Set $J = k/(2XY)$ and $a = km(1-X^2)/(2X^2Y) > 0$. At a minimum-cost contract in the benchmark $d = M_0$, let z_0 denote the optimal centered precision of the final-decision record. As $c_d \to \infty$, every such contract satisfies*

$$z_0 = \frac{a}{c_d} + O(c_d^{-2}),$$

$$T_0 - T_D = -\frac{a^2}{2c_d} + O(c_d^{-2}), \tag{27}$$

$$W_0 - W_D = \frac{amJ(3X^2-1)}{4Xc_d} + O(c_d^{-2}), \tag{28}$$

$$C_D - C_0 = \frac{amJ(1+X^2)}{4Xc_d} + O(c_d^{-2}). \tag{29}$$

Here T_D and W_D are evaluated at high effort. Thus $W_0 > W_D$ for sufficiently large finite prices when $X^2 > 1/3$, and $W_0 < W_D$ when $X^2 < 1/3$.

Proof. Write the benchmark objective excluding the final-decision acquisition payment as

$$F(x,y,z) = \frac{kh_H}{2} + \frac{k(1+mxz)}{2y(x+mz)} + \frac{c_0x^2 + c_\theta y^2}{2}.$$

Every global minimizer converges to $(X, Y, 0)$ as $c_d \to \infty$: comparison with T_D forces $z \to 0$, and compactness together with the unique direct optimum then forces $(x, y) \to (X, Y)$. Near that point the conditional (x, y) problem is smooth and strongly convex. Its optimum is an interior smooth function of z. Let H_D be the Hessian of the direct objective. At (X, Y),

$$H_D = J \begin{pmatrix} 3/X^2 & 1/(XY) \\ 1/(XY) & 3/Y^2 \end{pmatrix}, \qquad \gamma(x, y) = \partial_z F(x, y, 0) = -\frac{km(1 - x^2)}{2x^2 y}.$$

The conditional optimum therefore has expansion

$$\begin{pmatrix} x(z) \\ y(z) \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} - H_D^{-1} \nabla\gamma(X, Y) z + O(z^2).$$

Since $\gamma(X, Y) = -a$, minimization over z gives $z_0 = a/c_d + O(c_d^{-2})$ and the total-cost change in (27). The objective after minimizing over x and y is locally strictly convex for all sufficiently large c_d, so these expansions apply to every global optimum.

Let $W_D(x, y) = k[h_H + (xy)^{-1}]/2$ and $C_D(x, y) = (c_0 x^2 + c_\theta y^2)/2$ denote the expected wage and record acquisition cost functions for direct evaluation. All gradients below are evaluated at (X, Y). The first-order condition gives $\nabla W_D = -\nabla C_D$, with $\nabla C_D = J(1/X, 1/Y)$. Calculation gives

$$\nabla C_D^\intercal H_D^{-1} \nabla\gamma = \frac{mJ(3 - X^2)}{4X}.$$

The change in expected wages is consequently

$$W_0 - W_D = \left[\frac{mJ(3 - X^2)}{4X} - a\right] z_0 + O(z_0^2) = \frac{mJ(3X^2 - 1)}{4X} z_0 + O(z_0^2),$$

which yields (28). The acquisition change is

$$C_0 - C_D = -\frac{mJ(3 - X^2)}{4X} z_0 + \frac{c_d z_0^2}{2} + O(z_0^2).$$

Using $a = mJ(1 - X^2)/X$ proves (29). □

C.6 Pure effort under small observation noise

Lemma C.5. *In the model of Section 2, if $v\Phi(-1/\tau) < k\alpha_L$, then for every $r \in (1/2, 1)$,*

$$V^R(r) = \max\{vA_L(r),\ vA_H(r) - T_G(r)\}.$$

If $vA_H(r) - T_G(r) > vA_L(r)$, contracts guaranteeing high effort approach the robust value. If $r \geq \beta_H$, the contract with zero wages and no informative records is optimal.

Proof. At a target $q > \alpha_L$ with $\beta(q) \leq r$, the equilibrium accuracy bounds $A(q; r) \leq r$ and $A_L(r) \geq r - (2r - 1)\Phi(-1/\tau)$ give

$$v[A(q; r) - A_L(r)] \leq v(2r - 1)\Phi(-1/\tau) < k\alpha_L < kq.$$

Expected wages are at least kq, so zero wages give strictly higher profit. A contract with a low-effort equilibrium is also bounded by vA_L. If an interior equilibrium satisfies $\beta(q) > r$, the analyst is indifferent throughout the interval up to α_H where the manager follows M_0. Continuity extends that indifference to its lower boundary: either α_L, or a target with $\beta(q) = r$. Either equilibrium bounds the contract's robust value by vA_L. The remaining contracts guarantee high effort, and their profit supremum is $vA_H - T_G$. This proves the formula. For $r \geq \beta_H$, the first bound applies to every positive probability of high effort. □